User Manual
The Node Controller Suite

How to use the Node Controller web tools day to day: what every page, tab and button does, with a technical close to each chapter for when you want to know how it works. Written for the Pi Zero 2 W build; full-build differences are marked where they matter.

DASHBOARD SYSMON INSTMON WIFIMON 44HELPER

FILES COVERED BY THIS EDITION

TOOL	FILE	VERSION
Dashboard	nodecontrol_pi02w_v9_3_71_17_20261013.py	9.3.71.17-pi02w
SysMon	sysmon_pi02w_v6_13_67_15_20261013.py	6.13.67.15-pi02w
44helper	44helper_v0_0_183x_20261013.py	0.0.183x
instmon	instmon_v2_46_0_20261012.py	2.46.0
wifimon	wifimon_v5_30_20261012.py	5.30
Installer	install_all_v7_1_20261012.sh	7.1
SysMon FULL	sysmon_v6_13_81_20261013.py	6.13.81
Dashboard FULL	nodecontrol_v9_3_77_20261012.py	9.3.77

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How to use this manual

If you are new to the suite, read Part I first, then turn to the chapter for the tool in front of you. Every chapter after Chapter 2 follows the same order, so you always know where to look:

1. **What it's for**: what the tool or tab does, and when you would open it.
2. **Before you start**: what has to be installed, and anything that can go wrong.
3. **Tab by tab**: each card and button, and what good and bad readings look like.
4. **Common tasks**: short recipes, step by step.
5. **Under the hood**: the technical close, in a purple-labelled panel. It covers the files, services, ports and timers involved. You can skip it on a first read.

Markers used in the text

NOTE Background, or something worth knowing.

TIP A quicker or safer way to do something.

CAREFUL This changes your node, restarts something or needs a reboot.

FULL BUILD Applies to the full build (Pi 3, 4 or 5) only. Also shown as the badge **FULL**.

WORKING Used on a live node and working. Also shown as the badge **WORKING**.

TESTING Built and in use, but still being tested. Expect rough edges, and report what you find. Also shown as the badge **TESTING**.

Commands and pictures

Commands are typed on the Pi, in an SSH session or at a keyboard and screen. Lines starting with `#` are notes. Replace `<pi-address>` with your Pi's IP address or host name.

```
# a note about the next line
sudo systemctl status sysmon
```

Pictures marked **Illustration** are drawn to show a page's layout. They use the same tab names and order as the real page, but the values in them are made up. Example identities use the placeholders `N0CALL` and `<NODE>`.

PART I

Getting started

What the suite is, what each tool is for, how to log in, and the short version of installing and updating it. The full install steps are in the install manuals in the `docs/` folder.

1. The suite at a glance
2. Installing and updating, in brief

The suite at a glance

Node Controller is a set of five small web tools that run on the same Raspberry Pi as your AllStarLink node. Each tool has its own page in the browser. Together they let you run, check, repair and update the node without typing commands.

1.1 What is on the node

The tools sit on top of a working AllStarLink 3 (ASL3) node with DVSwitch. It helps to know the main parts, because the tools check and talk to all of them:

PART	WHAT IT DOES
Asterisk + app_rpt	The ASL3 engine. It runs your node, links to other AllStarLink nodes over IAX2 and handles EchoLink.
Analog_Bridge	Turns the node's audio into digital voice and back again. It talks to Asterisk on a private node (1999 by convention).
MMDVM_Bridge	Connects Analog_Bridge to the digital networks: DMR directly, and System Fusion, P25, NXDN and D-STAR through their gateways.
Gateways	YSFGateway, P25Gateway, NXDNGateway and ircDDBGateway. Each one connects to that mode's reflectors or talkgroups.
USRP2M17	Bridges a second private node (1917 by convention) to an M17 reflector.

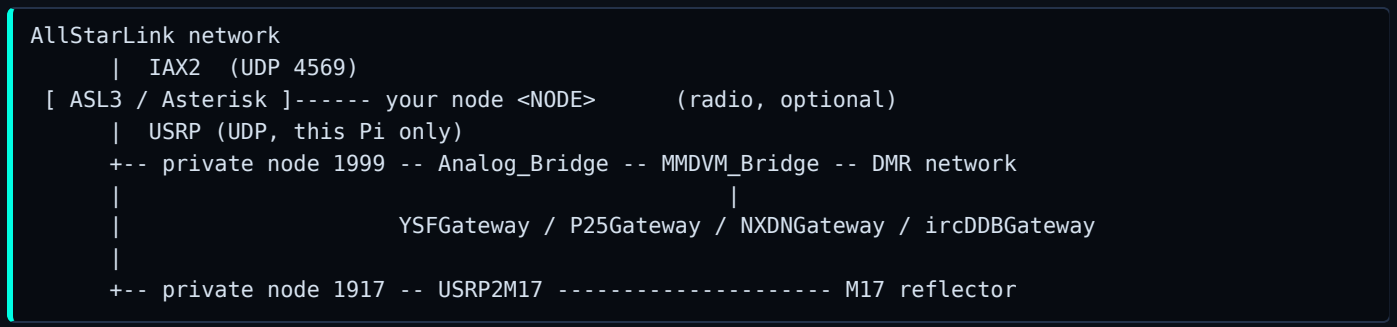


Figure 1.1 The audio path through a bridge node. Node numbers 1000 to 1999 are private: they never leave the Pi. The full layout is in `docs/ASL-DVS_Idealized_Node_Configuration.pdf`.

1.2 The five tools

Each tool is one file, runs as its own service, and serves its own page on its own port.

TOOL	PORT	WHAT IT'S FOR	OPEN IT WHEN YOU WANT TO...
Dashboard	8989	Node control.	Connect to AllStar or EchoLink nodes, change DMR talkgroup or YSF, P25, NXDN, D-STAR, XLX or M17 reflector, use favorites, make a phone call.
SysMon	9999	System monitor and health checks.	See whether the Pi and node are healthy, find a stopped service or a bad config file, read logs, look after the SD card.
44helper	9997	Setup, repair and configuration helper.	Set up 44Net or Cloudflare remote access, install ASL3, DVSwitch, SVXLink, M17 or Ampersand, run updates, save and restore your config.

instmon	8990	Installer and manager for the suite.	Update the tools from GitHub, start or stop them, pause the suite for a big job, back up or image the SD card.
wifimon	8991	WiFi and power watchdog.	Change WiFi networks, check signal and supply voltage, set up the automatic safe shutdown.

> **NOTE** 44helper is optional. The other four make up the core suite. A sixth item, the Watchdog script, is no longer needed: the Dashboard's own service restarts it if it stops.

Which tool for which job

I WANT TO...	GO TO
Talk on a DMR talkgroup or a reflector	Dashboard, the mode's tab
Link to another AllStar node	Dashboard, ASL tab
Find out why something stopped working	SysMon, Overview and Services, then the matching check tab
Check that my config files agree with each other	SysMon, ASL-DVS and DVSM tabs; 44helper, Config tab
Save my config before I change it	44helper, Config tab (restore point)
Reach my node from outside my home network	44helper, Router Install / Tunnel tabs or Cloudflare
Update the suite	instmon, GitHub Updates
Update Linux and ASL packages	44helper, Update tab
Make the SD card last longer	SysMon, SD Card tab; 44helper, System Optimization
Change WiFi network	wifimon

1.3 A first look at each page

Every page has a header with the tool's name and version and a Log Out button. Most pages have a row of tabs under the header; click a tab to switch view. The pictures below show each tool's tabs in order.

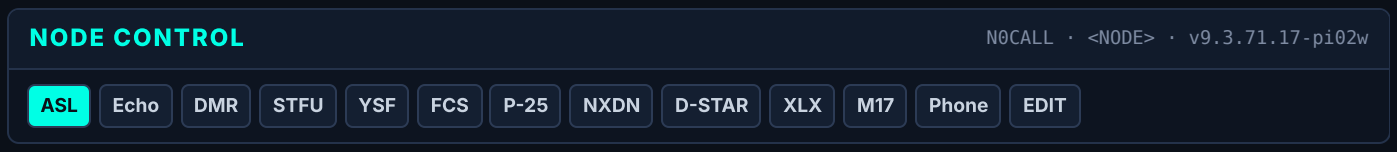


Figure 1.2 · Illustration. The Dashboard (port 8989). One tab per mode, plus Phone and EDIT. EDIT holds your favorites and settings. The Pi Zero 2 W build keeps every tab of the full build.

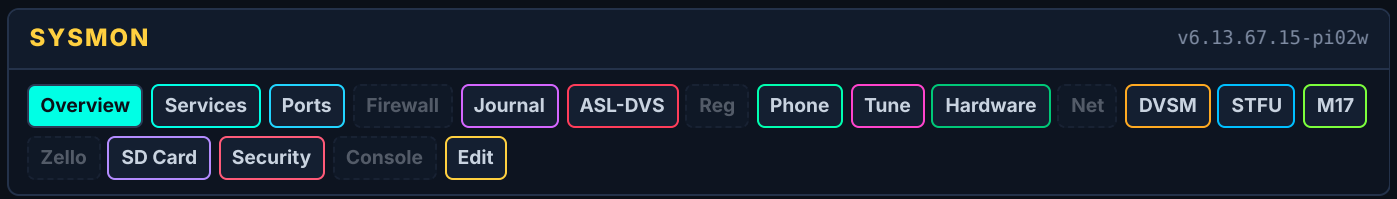


Figure 1.3 · Illustration. SysMon (port 9999). The 14 solid tabs are in the Pi Zero 2 W build. The 5 faded tabs (Firewall, Reg, Net, Zello, Console) are FULL build only.

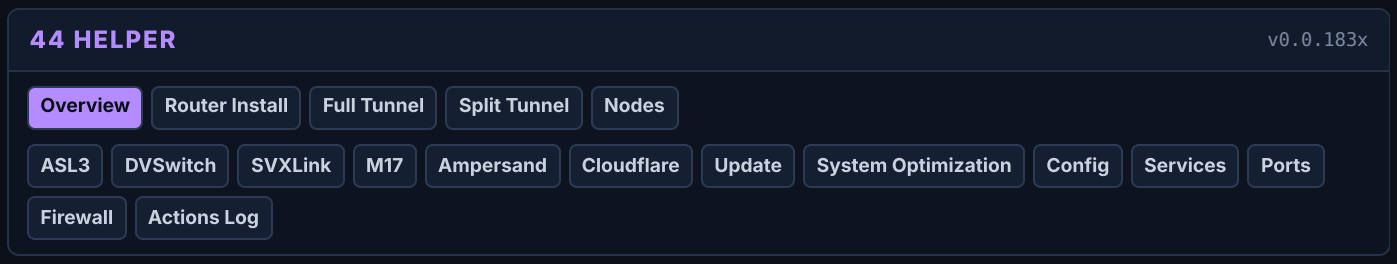


Figure 1.4 · Illustration. 44helper (port 9997). The first row is 44Net remote access. The second row has the installers, maintenance, Config and the status tabs. 44helper is the same on every Pi.

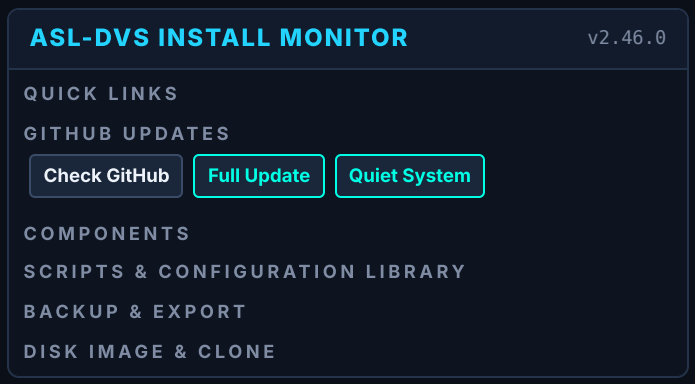


Figure 1.5 · Illustration. instmon (port 8990) is one long page of sections, not tabs.

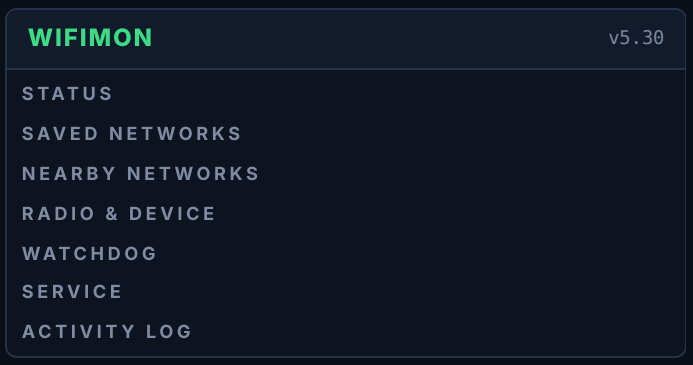


Figure 1.6 · Illustration. wifimon (port 8991) is also one page of cards.

1.4 Two builds: Pi Zero 2 W and full

The Dashboard and SysMon come in two builds. The other tools have a single build that runs on every Pi.

PI ZERO 2 W BUILD `PI02W`

For the 512 MB Pi Zero 2 W. This manual is written for it.

- Dashboard: every tab is kept. It searches the AllStar and EchoLink lists from disk instead of memory, and refreshes less often.
- SysMon: drops the Firewall, Reg, Net, Zello and Console tabs to save memory.

FULL BUILD `FULL`

For a Pi 3, 4 or 5.

- Dashboard: the same tabs, and it keeps the lists in memory.
- SysMon: all 19 tabs. Chapter 6 covers the five extra ones.

Which build am I running? Look at the version in the page header. A Pi Zero 2 W build version ends in `-pi02w`, for example `6.13.67.11-pi02w`. A full build version doesn't, for example `6.13.77`.

> **NOTE** Apart from SysMon, the builds look and work the same, so this manual points out the full build only where it differs.

1.5 Logging in

1. Open the page in a browser: `http://<pi-address>:<port>`, for example `http://192.168.1.50:9999` for SysMon.
2. Type the Pi's **root password** and press Log In. There is no user name.
3. When you're done, press **Log Out** in the header, especially on a shared computer.

- **The Dashboard and SysMon share one login.** Log in to either one and the other opens without asking, in the same browser. A shared login lasts 12 hours from the last time you used either page.
- **instmon, wifimon and 44helper each have their own login.**
- **Wrong passwords lock you out for a while.** On the Dashboard and SysMon, five wrong passwords within 5 minutes lock that address out of both for 15 minutes. The other tools have similar limits. Wait it out; don't keep guessing.
- **44helper stays logged in** until you log out, press **Log out all devices**, or the Pi reboots, because its jobs can run for a long time.

> **CAREFUL** The pages use plain HTTP, so your password crosses the network unencrypted. Only open them on your own network or over a VPN or tunnel, never straight from the internet. Chapters 7 and 8 cover safe remote access with 44helper.

> **TIP** If the login is always refused, set the root password again on the Pi with `sudo passwd root`. Raspberry Pi OS ships with root locked, so a new Pi needs this once.

1.6 Phones and tablets

The pages are laid out for phone, tablet and desktop widths. On a phone, wide tables hide their less important columns, and the tab row wraps onto more lines. Turning a tablet to landscape shows everything.

Leave a page open only while you're using it. Each open page polls its tool for fresh data, and on a Pi Zero 2 W that costs CPU and memory.

UNDER THE HOOD

Services, files and ports

TOOL	SYSTEMD SERVICE	PROGRAM	CONFIG
Dashboard	`nodecontrol`	`/usr/local/bin/nodecontrol.py`	`/etc/asl_dvs/asl_dvs.conf`
SysMon	`sysmon`	`/usr/local/bin/sysmon.py`	`/etc/sysmon/sysmon.conf`
44helper	`44helper`	`/usr/local/bin/44helper.py`	`/etc/44helper/44helper.conf`
instmon	`instmon`	`/usr/local/bin/instmon.py`	`/etc/asl_dvs/instmon_library/`
wifimon	`wifimon`	`/usr/local/bin/wifimon.py`	`/etc/wifimon/wifimon.conf`

Every program path is a link into a folder of its own under `/usr/local/lib/asl_dvs/` : `instmon/` (from instmon 2.40.0), `wifimon/` (wifimon 5.29), `44helper/` (44helper 0.0.180), `sysmon/` (SysMon 6.13.78 and 6.13.67.12-pi02w; `/usr/local/lib/asl_dvs_sysmon/` before, removed on install) and `nodecontrol/` (the Dashboard, from 9.3.76 and 9.3.71.15-pi02w; it was `dashboard/` in 9.3.75 and 9.3.71.14-pi02w, and directly in `/usr/local/lib/asl_dvs/` before that). From those versions the Dashboard is named **nodecontrol**: file `nodecontrol[_pi02w]_v….py`, service `nodecontrol`, program `/usr/local/bin/nodecontrol.py`. `/usr/local/bin/asl_dvs_dashboard.py` stays as a second link for older tools, and installing it removes the old `asl_dvs_dashboard` service. 44helper's file is `44helper_v….py` from 0.0.181, and the installer and uninstaller are `install_all_v….sh` and `uninstall_all_v….sh` from 7.1 and 1.5. instmon 2.43.0 knows both the old and new names. The installed file keeps its own name, so the version is always visible on disk.

One process each, started through the launcher

Every service starts its tool through the shared launcher `/usr/local/bin/asl_dvs_launch.py` :

```
ExecStart=/usr/bin/python3 /usr/local/bin/asl_dvs_launch.py /usr/local/bin/sysmon.py
```

The launcher loads the tool as a module so that Python keeps its compiled copy in `__pycache__`. That roughly halves each tool's memory (SysMon on a Pi Zero 2 W: 54 MB started directly, 26 MB through the launcher). The launcher also turns on three memory savers: `python3 -OO`, `MALLOC_ARENA_MAX=2` and a periodic `malloc_trim`. Each has an off-switch file in `/etc/asl_dvs/` named `launch_no_optimize`, `launch_no_arena_cap` or `launch_no_trim`.

How a page works

- Each tool is a small Python HTTP server. The page's HTML, CSS and JavaScript are built into the tool's file, so there is nothing else to install and no web server to configure.
- The page fetches live data from the tool's own `/api/...` addresses as JSON and redraws its cards.
- The login checks the password against the root entry in `/etc/shadow`, and sets a session cookie. Requests that change anything also need a matching CSRF header, so another web page can't act for you.
- Version numbers come from the `VERSION` line in each file. The `-pi02w` suffix is how the installer and instmon recognise the Pi Zero 2 W build.

Checking on a tool from the shell

```
# is it running, and its last 30 log lines
systemctl status sysmon --no-pager
journalctl -u sysmon -n 30 --no-pager
# is anything listening on its port?
ss -tlnp | grep 9999
# restart all the web tools
sudo systemctl restart sysmon nodecontrol instmon wifimon 44helper
```

Installing and updating, in brief

This chapter covers just enough to get the suite onto a node and keep it current. For every step and option, use the install manual for your build.

MANUAL	FILE IN DOCS/
Pi Zero 2 W install	ASL-DVS_Pi02w_Install_Manual.pdf
Full build install FULL	ASL-DVS_Full_Build_Install_Manual.pdf

2.1 Before you start

- A Raspberry Pi already running AllStarLink 3 with DVSwitch set up.
- SSH access with `sudo`, and a root password (`sudo passwd root`).
- On a Pi Zero 2 W, some free memory. Check with `free -m`. The install manual lists services you can turn off if you don't use them.

2.2 Install in three steps

1. **Get the files.** Download the current files from github.com/kd8pgk/ASL-DVS into one folder on the Pi.
2. **Install instmon first.** It can then install and update everything else from the browser.

   ```
   sudo python3 instmon_v2_46_0_20261012.py --install
   ```

   From 2.45.0 the install also downloads the manuals (PDF) from GitHub, so they are on instmon's Manuals card straight away. With no internet the install still finishes, and instmon downloads them once it can reach GitHub.
3. **Install the rest** with the installer. On a new node it asks for your callsign, node number and a label.

   ```
   sudo bash install_all_v7_1_20261012.sh
   ```

The installer picks the build for you: the Pi Zero 2 W build on a Pi Zero 2 W, the full build on anything else. If one build is already installed, it keeps that build. When it finishes, it checks that each tool is running and listening on its port.

> **TIP** 44helper is optional. Install it the same way, or from instmon's page, when you want its remote-access, installer or Config tools.

2.3 First visit to each page

PAGE	ADDRESS	CHECK FIRST
Dashboard	http://<pi-address>:8989	Your callsign and node number are right in the header.
SysMon	http://<pi-address>:9999	The Overview shows no red items (Chapter 3).
instmon	http://<pi-address>:8990	Every tool shows as installed and running.
wifimon	http://<pi-address>:8991	Answer "Turn the watchdogs on?" (below).

44helper	http://<pi-address>:9997	Take a restore point on the Config tab (Chapter 11).

wifimon's watchdogs

A new wifimon starts with both watchdogs off: it shows WiFi and voltage, but won't act on them. On your first login it asks whether to turn them on. **Add your WiFi networks first.**

- **WiFi watchdog:** reconnects to your saved networks, and shuts the Pi down after 3 minutes with no connection.
- **Voltage watchdog:** shuts the Pi down after 30 seconds of low supply voltage.

> **CAREFUL** Both watchdogs can shut the node down. A Pi Zero 2 W is sensitive to voltage drop, so use a good power supply and a short cable before turning the voltage watchdog on.

2.4 Keeping the suite up to date

Updates come from GitHub through instmon's **GitHub Updates** section.

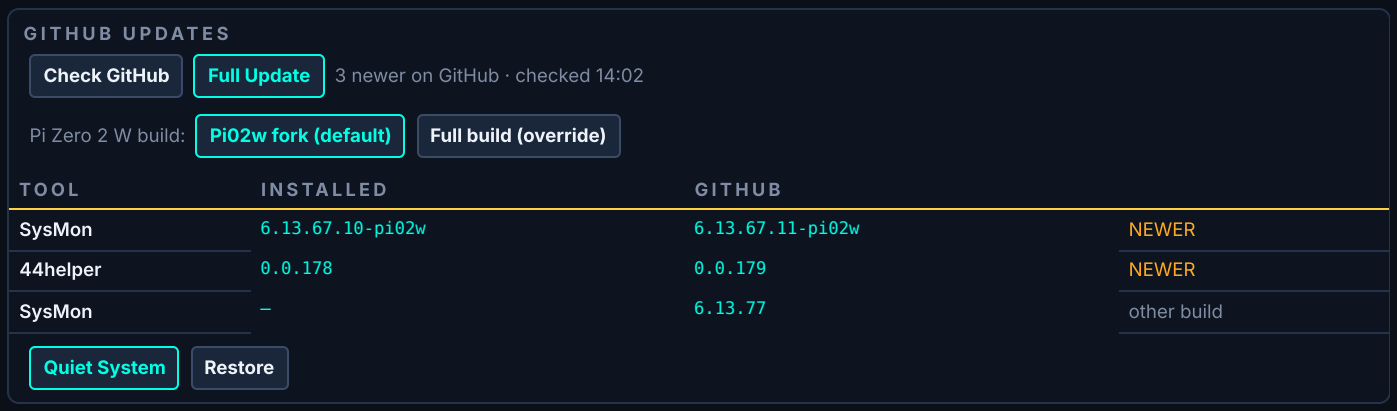


Figure 2.1 · Illustration. instmon's GitHub Updates section on a Pi Zero 2 W. The build choice appears only on a Pi Zero 2 W.

BUTTON	WHAT IT DOES
Check GitHub	Lists the newest copy of each tool on GitHub, and marks the ones newer than yours.
Update	Downloads and checks one file into instmon's library. Nothing is installed until you press Install on it.
Full Update	Updates every installed tool in one go, one tool at a time, with automatic roll-back if a new version fails.
Quiet System	Pauses SysMon, the Dashboard and 44helper during a big upload or install. The radio side keeps running, and everything comes back by itself after 30 minutes. **Restore** brings it back sooner.

> **TIP** Before a Full Update, take a restore point in 44helper's Config tab. Full Update only replaces the tools, but a restore point means you can roll back any config change you make while trying out the new versions.

Updating Linux, ASL3 and DVSwitch packages is a separate job. 44helper's Update tab handles it (Chapter 10).

2.5 Removing the suite

```
# remove one tool
sudo python3 /usr/local/bin/sysmon.py --uninstall
# remove every tool except instmon
sudo bash uninstall_all_v1_5.sh
```

Your settings in `/etc/asl_dvs/`, instmon's library and wifimon's config stay in place. Remove instmon last, with its own `--uninstall`.

UNDER THE HOOD

What install and update actually do

--install, per tool

- Copies the file into place, writes its `.service` unit, and writes the shared launcher if it is missing.
- Enables the service so it starts at boot, and starts it.
- `--uninstall` stops and removes the service and the program. It removes the launcher only once no other service uses it.

How the installer picks a build

1. `SYSMON_VARIANT` / `DASH_VARIANT` set to `pi02w` or `full`, if you set them;
2. otherwise, the build already installed;
3. otherwise, `pi02w` on a Pi Zero 2 W and `full` on any other Pi.

If no file of the chosen build is found, that tool is skipped with a warning; the other build is never used in its place. The installer looks in its own folder, the current folder, `/root`, `/home/pi`, your home folder, `/tmp` and instmon's library. `--non-interactive` (or `-y`) skips every question and takes the identity from `AUTO_CALLSIGN`, `AUTO_NODE` and `AUTO_LABEL`.

The checks on every download

Before instmon saves a file from GitHub, the file must match GitHub's file list. It must also have a version line and the right first line, pass a syntax check, and support `--install`.

Full Update, step by step

1. Downloads and checks every new file before anything changes.
2. Downloads the manuals that are new or changed on GitHub (from 2.45.0), except ones you deleted (from 2.46.0). A failure here is listed as skipped and the tools still update.
3. Pauses the web tools.
4. For each tool in turn, saves a copy of the installed file in the library, runs the old `--uninstall`, and runs the new `--install`. If the new one fails, it puts the old one back and moves on.
5. Updates instmon last. A timer reinstalls the old instmon if the new one isn't answering after 90 seconds.
6. Ends with a summary, including any manuals that were downloaded. New install scripts go only to the Scripts library and are never run for you.

On a Pi Zero 2 W, the build choice decides which build Full Update installs. A tool installed in the other build is listed as **SWITCH** and replaced. On other Pis the choice is hidden and each tool keeps its build.

Settings that survive an update

Config files in `/etc` are never touched by an update. SysMon's Hardware tab options (HDMI and video driver off, lower GPU memory) are saved when SysMon is uninstalled during an update and put back on install, if the install follows within 30 minutes.

PART II

SysMon

SysMon is the node's health monitor. It shows whether every service is up, checks your Asterisk, DVSwitch and M17 config files against each other, reads logs, and looks after the hardware and SD card. Most of what it does is read-only. Every button that changes something says so.

3. SysMon basics and live status
4. Node and DVSwitch health
5. System care
6. Full-build tabs

SysMon basics and live status

This chapter covers the parts of SysMon you see every time: the header and command bar, then the four status tabs: Overview, Services, Ports and Journal. Open SysMon at `http://<pi-address>:9999`.

3.1 What SysMon is for

Open SysMon when something on the node isn't working, or when you want to be sure everything is. In one place it shows:

- whether Asterisk, the DVSwitch bridges, the gateways, USRP2M17 and the Node Controller tools are running;
- the Pi's load, memory, temperature and power health, updated every 5 seconds;
- which programs are listening on which network ports, and whether two of them clash;
- the system log (journal) of any service, filtered and ready to copy.

The status tabs can also start, stop and restart services. Everything else in SysMon is either read-only or asks before it changes a file.

3.2 The page at a glance

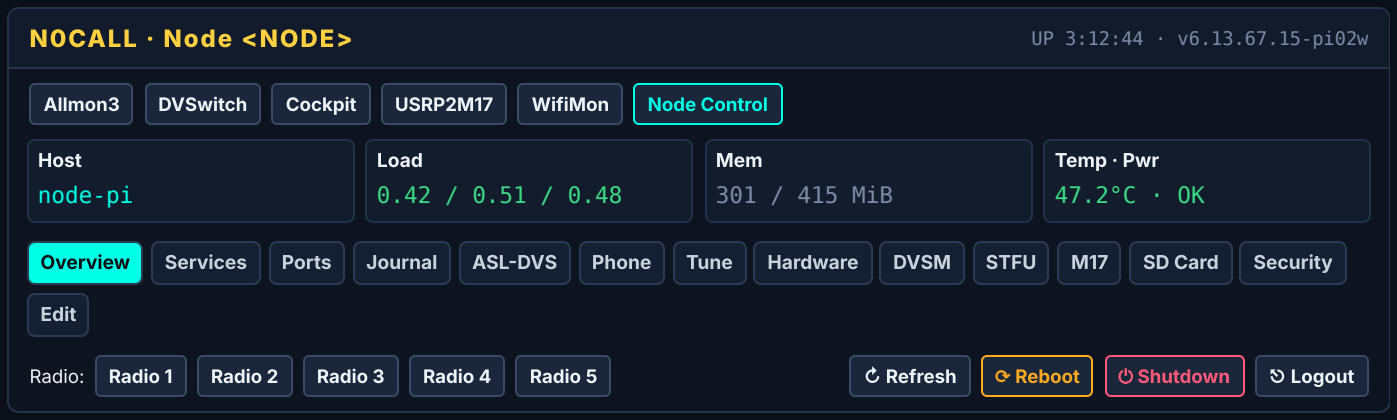


Figure 3.1 · Illustration. SysMon's top section. From the top: header, quick links, status strip, tabs and command bar.

PART	WHAT IT SHOWS OR DOES
Header	Your callsign and node number, how long the Pi has been up, and SysMon's version.
Quick links	Open Allmon3, the DVSwitch page, Cockpit, the USRP2M17 page and wifimon in a new tab. **Node Control** goes back to the Dashboard.
Status strip	Host name, load, memory, CPU temperature and power health (below).
Tabs	One per view. You can hide the ones you don't use on the Edit tab (Chapter 5).
Radio 1 to 5	Saved sets of radio audio settings. Pressing one applies it (Chapter 5, Tune tab).
Refresh	Reads the status strip again now, instead of waiting for the next 5-second update.
Reboot / Shutdown	Restart or power off the whole Pi. Both ask first. After Shutdown, the Pi stays off until you remove and restore power.

Reading the status strip

FIELD	GREEN	AMBER	RED
Load (1, 5, 15 min)	1-minute load up to 1.5	above 1.5	above 3
Temp	up to 60 °C	above 60 °C	above 75 °C
Pwr	`OK`	`WARN` : a problem happened since boot but isn't happening now	`⚡ UV` under-voltage now, or `⚡ THR` throttled now

Hover over **Pwr** to see exactly what was flagged, such as "Under-voltage since boot". When under-voltage starts, SysMon also shows a pop-up warning to check the power supply.

> **TIP** A Pi Zero 2 W has four 1 GHz cores and 512 MB, so a load of 1 to 2 during a tune or an update is normal. A load that *stays* red with nothing going on usually means a service is stuck restarting. Check the Overview for an `NR:` count.

The status dots

Every service list in SysMon uses the same coloured dots:

DOT	MEANS
● **Active**	Running normally.
● **Stopped**	Installed but not running. That is fine for a gateway you don't use, and a problem for Asterisk.
● **Failed / Masked**	It crashed or failed to start, or it has been blocked from starting (masked).
● **Unknown**	SysMon couldn't read its state.
● Not installed	There is no such service on this Pi. That is normal for optional parts.

Two banners you may see

- **"Restart sysmon to apply server address changes"**: you changed SysMon's port or bind address on the Edit tab. It takes effect when SysMon restarts.
- **"wifimon — Node Shutting Down"**: wifimon has started a safe shutdown because of low voltage or lost network. The Pi needs power-cycling, or its WiFi back, before it will come up again.

3.3 Overview tab

The Overview is SysMon's home page. It lists your **pinned services** in groups. Out of the box the groups are System, ASL-DVS, Asterisk Core and M17, which cover everything a bridge node runs.

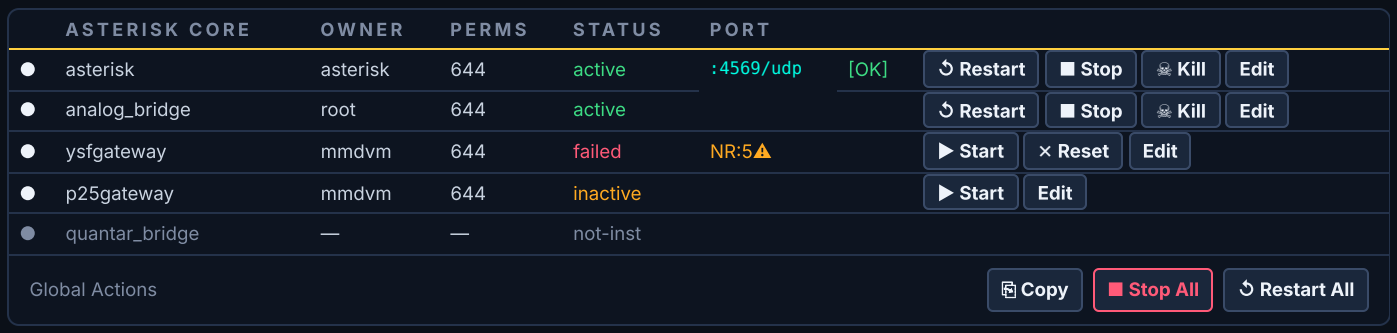


Figure 3.2 · Illustration. One Overview group. The buttons on each row depend on the service's state.

COLUMN	WHAT IT TELLS YOU
Name	The service. Click it to open its detail panel (3.4).
Owner	The Linux user it runs as. Amber means it runs as root but doesn't need to. Click the amber value for the Security tab's explanation.
Perms	The permissions on its service file. Amber or red means other users could change it. Click for the explanation.
Status	`active`, `inactive`, `failed`, `not-inst` or `unknown`.
Port	The port it should listen on. If it has restarted on its own since it was last started, this shows `NR:` and the count instead, with ⚠ above 3.
[OK] / [DUP]	For a running service with a port: `[DUP]` means another program has the same port and protocol, and hovering names both. `[OK]` means no clash.

Row buttons

STATE	BUTTONS
active	**Restart**; **Stop** (asks first); **Kill**, which ends the process at once with no clean shutdown (asks first); **Edit**, which opens the service file.
failed	**Start**; **Reset**, which clears the failed mark and the restart counter; **Edit**.
inactive	**Start**; **Edit**.

Restart is done carefully: SysMon stops the service, waits for it to stop, starts it, and waits for it to come up. The Overview refreshes itself a few times over the next half minute while that happens.

Global Actions

- **Copy** puts the whole Overview on the clipboard as text, ready to paste into a help request.
- **Stop All** stops every pinned service except SysMon itself.
- **Restart All** restarts every pinned service, SysMon included.

> **CAREFUL** **Stop All** and **Restart All** act at once, with no confirmation. Stop All takes the node off the air: it stops Asterisk, the bridges and the Dashboard. To get it back, press **Restart All** or reboot. SysMon keeps running, so the page stays usable.

3.4 Services tab

The Services tab lists **every** service on the Pi, not just the pinned ones.

- **Active** shows only running services. **All** also shows stopped, failed and disabled ones.
- Type in **filter by name** to narrow the list, for example `gateway`.
- The list refreshes every 30 seconds while the tab is open. **Copy** puts it on the clipboard as text.

The service detail panel

Click any service, here or on the Overview, to open its panel. The top shows badges for its state, whether it starts at boot, its process number, its restart count (`NR:`), CPU and memory use, and the user it runs as. Under that, **Ownership & permissions** shows who owns the service file and the program.

SECTION	BUTTON	WHAT IT DOES, IN PLAIN WORDS
Daemon Controls	Reload Daemon	Makes systemd re-read every service file. Needed after you edit one by hand.
	Reset Failed	Clears a "failed" mark so the service can be started cleanly.
	Mask / Unmask	Mask blocks a service from starting at all, even by hand, until you Unmask it. Asks first.
Runtime Controls	Start / Stop / Restart	As on the Overview. Stop asks first.
	Reload	Asks the program to re-read its settings without restarting. Greyed out for services that don't support it.
Boot Persistence	Enable / Disable	Whether the service starts by itself at boot. Disable asks first. Neither one starts or stops it now.
Pinned Services	Pin / Unpin	Adds the service to the Overview, in a group you pick or a new one, or removes it.
Status	(output)	The service's current status and its last 50 log lines, with errors in red and warnings in amber.

Editing a service file

Edit opens the service's systemd file in an editor, with **Save**, **Restart**, **Reload Daemon** and **Copy** buttons. After saving, press Reload Daemon and then Restart for the change to take effect.

> **NOTE** Service files that came with a package live in `/lib/systemd/system`. SysMon never changes those. Saving one creates your own copy in `/etc/systemd/system/`, which takes priority and survives package updates. A yellow bar warns you when this happens.

3.5 Ports tab

The Ports tab lists every network port a program on the Pi is listening on. It refreshes every 15 seconds.

COLUMN	MEANING
Dot	Green: the port belongs to a known service. Grey: a listener SysMon can't tie to a service.
Port / Proto	The port number, and TCP or UDP.
Process / PID	The program listening, and its process number.
Known Service	The service that program belongs to, or "— unknown —".
Conflict	`[OK]`, or `[DUP]` if more than one program has the port.

- **All / TCP / UDP** filter the list. **Copy table** copies what you see; **Copy all** copies every port.
- Click a row for its panel: the listen address, **Probe** buttons (HTTP and TCP for a TCP port, UDP for a UDP port) that test whether it answers, and for a known service **Restart Service**, **Stop Service** and **Journal**.

> **NOTE** `[OK]` only means nothing else is fighting for that port. It doesn't mean the service is healthy. A DVSwitch port clash is a common cause of "no audio". The pairing rules are in `docs/ASL-DVS_Idealized_Node_Configuration.pdf`, section 4.

3.6 Journal tab

The Journal tab lists your pinned services that are installed, followed by any other running service. Click one to open its log in a pop-up window.

CONTROL	WHAT IT DOES
Lines 50 / 200 / All	How far back to read. All means the last 2,000 lines.
grep filter	Shows only lines containing your text, for example `error` or a talkgroup number.
Refresh	Reads the log again.
Restart / Start	Restart or start the service from the log window, so you can watch what happens.
Copy	Copies the lines you see (after the filter).

3.7 Common tasks

A service shows red

1. On the Overview, note the `NR:` count. A high count means it keeps crashing.
2. Open the Journal tab, click the service, choose 200 lines and filter on `error`.
3. Fix the cause. It is usually a config file, covered on the ASL-DVS, DVSM, STFU and M17 tabs (Chapter 4).
4. On the Overview, press **Reset**, then **Start**. Watch the dot turn green.

Find out what is using a port

1. Open the Ports tab and press UDP or TCP.
2. Find the port number. The Process and Known Service columns name the owner.
3. If it shows `[DUP]`, hover over it for both programs, then stop or reconfigure one of them.

Add a service to the Overview

1. Services tab, press **All**, and filter for the service.
2. Click it, pick a group under Pinned Services (or "New group...") and press **Pin**.

Send someone a status report

Use **Copy** on the Overview, the Services list, a Ports table or a journal window, and paste it into your message. Copies are plain text, so they paste cleanly into an email or forum post.

UNDER THE HOOD

How SysMon gathers and acts

Refresh timings

WHAT	HOW OFTEN	NOTES
Background poll (server)	every 10 s	Host, load, memory, swap, temperature, voltage, throttle state, disk, SD health, process count, wifimon shutdown state. Also tells systemd's watchdog that SysMon is alive.
Quick tests, port clashes	every 30 s	Every third background poll.
Status strip (page)	every 5 s	`GET /api/status`, read from the background poll's snapshot, so it costs almost nothing.
Services list	every 30 s	Only while the Services tab is open.
Ports list	every 15 s	Only while the Ports tab is open.
M17 reflector list	hourly check	Downloads `M17Hosts.json` again when it is stale (Chapter 4).

Switching tabs stops the previous tab's timer, so only the tab you're looking at polls.

Where the data comes from

- **Services:** `systemctl show` and `systemctl list-units`. The owner comes from the running process, or from `User=` in the unit.
- **Ports:** `ss -tlnpu`, with each process's command line from `/proc/<pid>/cmdline`.
- **Journal:** `journalctl -u <unit> -n <lines>`, between 10 and 2,000 lines.
- **Power:** `vcgencmd get_throttled`. Its bits give under-voltage, frequency cap, throttling and temperature limit, both now and since boot. Without it, SysMon shows the core voltage instead.

Service actions

Every action goes to `POST /api/svc`. Unit names are checked before anything runs.

- **Simple actions** (start, stop, reset-failed, enable, disable, reload, mask, unmask) run the matching `systemctl` command, with a 15-second limit.
- **Restart** runs in the background: `systemctl stop`, wait up to 15 s; if it is still running, `SIGKILL` and wait up to 5 s; then `systemctl start` and wait up to 10 s for it to become active. The result goes to SysMon's own log.
- **Kill** sends `SIGKILL` to the main process. systemd may then restart it, depending on the unit's `Restart=` setting.
- **Stop All** runs `systemctl stop` on each pinned unit except `sysmon.service`. **Restart All** runs `systemctl restart` on each pinned unit, SysMon included.

The pinned list

The Overview groups come from `[services] pinned` in `/etc/sysmon/sysmon.conf`, one line per entry:

```
__GROUP__:Asterisk Core
asterisk.service | ASL3 Asterisk/app_rpt | /etc/asterisk/rpt.conf | 4569 | udp
# unit | description | config_path | port | proto   ( - for none )
```

You can edit it as text on the Edit tab (Chapter 5). Pin and Unpin rewrite it for you.

Login and process

- SysMon runs as root, because it reads every config file and controls services. It refuses to start as any other user.
- SysMon and the Dashboard keep their logins in one shared file, `/run/asl_dvs/auth_session.json` (mode 600, locked while written). Both use the cookie `asl_dvs_session`, so one login covers both pages. A session expires 12 hours after it was last used. Five failures from one address within 5 minutes lock it out of both for 15 minutes. `/run` is cleared at reboot, so a reboot logs everyone out.
- The password is checked through PAM, with `/etc/shadow` as a fallback.
- The page is built once at start-up and sent compressed. Everything after that is small JSON requests.

```
# SysMon's own checks, without the web page
sudo python3 /usr/local/bin/sysmon.py --selftest
```

Node and DVSwitch health

Five SysMon tabs check the radio side of the node: ASL-DVS (the Asterisk, Allmon3 and DVSwitch config files), DVSM (DVSwitch Mobile accounts), STFU (BrandMeister), M17 (the USRP2M17 bridge) and Phone (autopatch). They read your files and tell you, in plain words, what is set, what is missing and how to fix it.

4.1 How the check tabs read

All five tabs use the same badges, so learn them once:

BADGE	MEANS	WHAT TO DO
PASS / OK	Set, and set to a real value rather than a default placeholder.	Nothing.
WARN	Works, but something is off or optional and missing.	Read the fix line; often it is fine to leave.
FAIL	Missing, still at a default such as `N0CALL`, or wrong.	Follow the fix line under it.
INFO	A value for your reference, such as a port or talkgroup.	Compare it with the other side of the link.
NONE	Listed but turned off (for example a module set to `noload`).	Nothing, unless you need it.

Many check lines end with a **fix**: the exact setting to add or the command to run. Copy buttons on each card give you the whole card as text.

> **CAREFUL** The **Edit** buttons on these tabs open the real config file, and **Save** overwrites it after one confirmation. SysMon writes the file safely (a power cut can't leave half a file), but it **does not keep a backup copy** of these files. Take a restore point in 44helper's Config tab first (Chapter 11), or press Copy before you change anything.

4.2 ASL-DVS tab

Four cards, one per group of files. Click a file row, or its **Edit** button, to open it in the editor.

/etc/asterisk/ — Config Files

Every `.conf` file in `/etc/asterisk`. The ones that matter most are checked under their name:

FILE	CHECKED
rpt.conf	Callsign is set and not a placeholder; each node number; its `rxchannel`, and for USRP nodes the RX and TX ports, with a reminder that Analog_Bridge must use the same pair.
manager.conf	At least one AMI user, and that each has a password.
echolink.conf	Callsign, password, EchoLink node number and the ASL node it links to.
rpt_http_registrations.conf	Each node that registers with AllStarLink, and that it has a password.
savenode.conf	Node and password pairs.
modules.conf	Which `chan_` channel drivers load, are required, or are set to `noload`. This list is folded away: press ▶ **Show**.

extensions.conf	The area code (`HOMENPA`, not the 999 default) and node number in `[globals]`.

Lines in a file that set a port get an `[OK]` or `[DUP]` badge, the same clash check as the Ports tab.

Allmon3 — Config Files

`allmon3.ini`, `web.ini`, `users`, `menu.ini` and `custom.css` in `/etc/allmon3`. `users` is read-only here: change Allmon3 logins with `allmon3-passwd`.

DVSwitch — Config Files

The DVSwitch files that exist on this Pi, from either their `/opt` or `/etc` location:

FILE	CHECKED
Analog_Bridge.ini	DMR ID and repeater ID (not defaults), AMBE mode, TX talkgroup and slot (1 or 2), colour code, software AMBE fallback, and the USRP RX and TX ports.
MMDVM_Bridge.ini	Callsign and DMR ID, location and description, and which modes (DMR, Fusion, P25, NXDN, D-STAR) are enabled, with their addresses. FAIL if no mode is enabled.
DVSwitch.ini	For each mode section: address and ports. For `[STFU]`, also the BrandMeister server, password, DMR ID and start talkgroup.
var.txt	The DMR networks the `dvs` menu knows about (BrandMeister, TGIF, DMR+, Other 1 and 2), and their addresses. Passwords are never shown.
Analog_Reflector.ini	If present: callsign, node ID and the USRP port.

D-Star — ircDDBGateway

Checks `/etc/ircddbgateway` for what the Dashboard's D-STAR gateway linking needs:

- the `ircddbgatewayd` service is running;
- `gatewayCallsign` is set, and `repeaterBand1` is a module letter. The card shows the 8-character name the Dashboard uses for the node, such as `N0CALL B`;
- ircDDB is on (`ircddbEnabled`, or one of `ircddbEnabled2`–`4`). WARN when it's off: reflectors still work, but gateway callsigns can't be found;
- `dextraEnabled` is on. FAIL when it's off: ircDDBGateway refuses gateway links and XRF reflectors without it;
- a setting left out of the file counts as ircDDBGateway's own default: `ircddbEnabled`, `ircddbEnabled2` and `dextraEnabled` are on unless the file turns them off;
- remote control (optional, WARN when off): when `remoteEnabled=1`, a password is set, the port is valid and something is listening on it. With it, the Dashboard hears whether a gateway link was accepted;
- `gatewayAddress` is blank or one of this Pi's own addresses, never `127.x` and never the router's address.

Buttons: **Edit /etc/ircddbgateway**, **Restart ircddbgatewayd** (asks first; any D-STAR link drops), **Refresh** and **Copy**.

4.3 DVSM tab

DVSwitch Mobile (DVSM) is the phone app for talking through your node. This tab reads your files and shows exactly what to type into each of the app's four account types. The **Node IP** in the tab header is this Pi's address on your network.

CARD	USE IT FOR	KEY SETTINGS SHOWN
IAX2 Direct	Phone-mode connection straight to your node.	Host (LAN and WAN), port, user name and password from `iax.conf`, CallerID, node. "Phone Mode IAX2" ticked.
IAXRpt	**Recommended.** Radio mode: least data, and it only sends audio while the channel is active.	As above, but Caller Num must be `0` and "Phone Mode IAX2" unticked.
Node Mode	The app registers as its own AllStarLink node.	register.allstarlink.org, port 4569, and a **spare** node number registered to your callsign. Auto Connect unticked.
USRP / DVSwitch Bridge	Connect to Analog_Bridge's USRP port.	Host (local, LAN or WAN), the TX and RX ports, callsign and DMR ID.

Passwords show as `••••••••`. Click the eye to see one while you copy it into the app.

Compatibility Checks

SETTING (RPT.CONF NODE)	WHAT IT DOES	SHOWN AS
propagate_dtmf	Lets DTMF commands from DVSM reach the node.	INFO if off: only needed for DVSM.
phonesendlinks	Fills the DVSM Status tab's linked-node list.	INFO if off.
propagate_phonedtmf	Passes phone callers' tones to every linked node, bridges included.	**WARN if on.** Leave it off unless you need it.
remote_dtmf_allowed	Lets any linked node send commands to this node.	**WARN if on.**

It also checks `iax.conf`: that an `[iaxrpt]` (or callsign-named) user with a password exists, which IAXRpt needs, and how call tokens are set.

4.4 STFU tab

STFU is DVSwitch's BrandMeister terminal: it connects the node to BrandMeister directly, as a hotspot. Its settings live in the `[STFU]` section of `DVSwitch.ini`.

CARD	SHOWS
Installation	Whether the STFU program and its service are installed, running and enabled at boot.
Configuration	BrandMeister server, port (default 62031) and hotspot password (masked); your DMR ID; the USRP ports; start talkgroup; subscriber file; talker alias; log level. A link opens BrandMeister's master-server list.
Compatibility Checks	Server set, password set, DMR ID not a default, ports set, subscriber file found (it turns DMR IDs into callsigns), service running.
Sample [STFU] Stanza	A complete section built from your values, with placeholders for anything missing. **Copy Stanza** or **Edit Stanza**.
Edit DVSwitch.ini	Expand to edit the whole file. **Save**, then **Restart STFU** to apply.

> **TIP** The BrandMeister hotspot password is not your BrandMeister login password. Set it on brandmeister.network under **Self Care → Hotspot Security**.

4.5 M17 tab

The M17 bridge, USRP2M17, links private node 1917 to an M17 reflector. The Dashboard's M17 tab rewrites its settings file every time you connect or disconnect.

CARD	SHOWS
Installation	The program, its service and whether it is running. **Restart usrp2m17** asks first.
Configuration	`/opt/USRP2M17/USRP2M17.ini` : callsign; the reflector name and address (`0.0.0.0` means not connected); the M17 and USRP ports; gain; logging.
Compatibility Checks	Callsign set; reflector connected (WARN when idle, which is normal); USRP ports are 32008 and 34008; service running; `rpt.conf` has a `[1917]` node whose `rxchannel` ports match the ini file.
Reflector Host File	`/var/lib/mmdvm/M17Hosts.json` : how many reflectors it lists and how old it is. **Update Now** downloads a fresh copy.
Sample USRP2M17.ini	A complete file built from your values.
Edit USRP2M17.ini	The real file. Save, then Restart from the Installation card.

CAREFUL Edits to `USRP2M17.ini` don't survive the next M17 connect or disconnect on the Dashboard, which rewrites the whole file. Make lasting M17 changes on the Dashboard instead.

NOTE If `LocalPort` under `[M17 Network]` is missing, the sample fills in `17010` , the port the Dashboard writes (Pi02w from 9.3.71.11, full build from 9.3.74; see the node configuration guide, section 4). SysMon before 6.13.67.10 (Pi02w) and 6.13.77 (full) filled in `32010` , which is the port MMDVM_Bridge uses for P25.

4.6 Phone tab

The Phone tab shows what Asterisk is set up to do for **autopatch** (radio dials out) and **reverse autopatch** (phone calls in to the radio). It is **read-only**. You set phone networks up on the Dashboard's Edit page, under Phone. The one exception is the Autopatch modules card.

CARD	WHAT IT TELLS YOU
Live	IDLE, DIALING, INCOMING or IN CALL; whether the phone node is linked to your node (it links while the Dashboard Phone tab is open); whether that tab is open (callers get busy when it isn't); which network is in use; phone patch on or off; calls now; sign-in state. Refreshes every 5 seconds.
Autopatch node	The private node the Dashboard made for each phone network.
Phone networks	Each network and its dialing target.
Hams Over IP TESTING	Shown only when you have a Hams Over IP account: the account and its sign-in.
Dialing rules	The dialplan contexts Asterisk has for each network.

Health checks	Asterisk running; nodes, files, live state and call routes all consistent; the three helper scripts (`*61` dial, `*62` / `*65` hang-up, `*980`–`*982` tones) present and runnable. Each check is OK, WARN, FAIL or INFO, with a fix.
Router test	Three tests for calls that won't connect or have one-way audio (below).
Autopatch modules	The Asterisk modules each feature needs, and whether each is loaded. You can change them here (below).
Recent call attempts	Up to 20 phone calls from the last 24 hours, in or out, with the network and the last 4 digits only.

Router test

1. **Going out: Run outgoing test** signs in to your phone provider once and sends one STUN packet. It takes about 5 seconds.
2. **Call audio:** start a **Test call** on the Dashboard's Phone tab, then press **Check audio now**. SysMon counts audio packets in and out over 4 seconds (SIP calls only). "Out but none in" points to the router forward or the Pi firewall. Audio in but none out is normal on a simplex node while nobody talks.
3. **Coming in:** press **Start watching (10 min)**, have someone call your number or extension, then press **Check**.

Autopatch modules

Modules are grouped by feature: **Autopatch** and **Reverse autopatch**, each READY, MISSING, OFF, NOT SET UP or ASTERISK DOWN. Modules both features need, and the ones every node needs, are locked (🔒).

- **Activate / Deactivate** a feature: sets every module it needs to load and starts it, or sets the modules only it uses to noload and stops them. The confirmation lists exactly which modules.
- Per module: **Load** / **No load** change `modules.conf` for the next Asterisk restart; **Start** / **Stop** load or unload it now.
- If a module file is missing from the Pi, the card names the package it comes from and offers **Reinstall**, with the same commands shown for doing it by hand.

> **NOTE** Unlike the file editors, this card backs up `modules.conf` before every change. It keeps the last 5 copies next to the file.

The full story of the phone dialplan is in `docs/ASL-DVS_Pi02w_Phone_Dialplan_Guide.pdf`.

4.7 Common tasks

No audio between AllStar and DMR

1. ASL-DVS tab: under `rpt.conf`, note node 1999's RX and TX ports.
2. Under `Analog_Bridge.ini`, check that its RX Port and TX Port are the same pair the other way round, as the note says.
3. Ports tab: check that neither port shows `[DUP]`.
4. Fix the file with Edit, then restart `analog_bridge` and `asterisk` on the Overview.

Set up DVSwitch Mobile

1. DVSM tab: make sure the IAXRpt card shows PASS. If not, follow the Compatibility Checks fix.

2. In the app, add an IAX2 account and copy each field from the IAXRpt card. Use Host (LAN) at home and Host (WAN) away.
3. Untick Phone Mode IAX2 and set Caller Num to 0.

Get a new node ready for STFU

1. STFU tab: work down the Compatibility Checks until every line shows PASS.
2. Copy the Sample [STFU] Stanza if your file has no `[STFU]` section, and paste it in with Edit DVSwitch.ini.
3. Save, then Restart STFU.

UNDER THE HOOD

Files, endpoints and what is sent to the page

Where each tab reads

TAB	FILES AND COMMANDS	ENDPOINT
ASL-DVS	every `*.conf` in `/etc/asterisk`; `/etc/allmon3/*`; the DVSwitch files under `/opt`, `/etc` and `/var/lib/dvswitch/dvs/var.txt`; `/etc/ircddbgateway`, plus `ss -Hlunp` and `ip -o -4 addr`	`/api/asterisk/files`, `/api/allmon3/files`, `/api/dvswitch/files`, `/api/dstar-gw`
DVSM	`iax.conf`, `rpt.conf`, Analog_Bridge's ini; the LAN address from the default route	`/api/dvsm`
STFU	`DVSwitch.ini [STFU]`, `/opt/STFU`, `stfu.service`	`/api/stfu`
M17	`/opt/USRP2M17/USRP2M17.ini`, `rpt.conf [1917]`, `usrp2m17.service`, `M17Hosts.json`	`/api/m17`
Phone	Asterisk's files and the CLI (`asterisk -rx`), `/var/log/asterisk/full` or `messages` (last 2 MB), or the journal	`/api/phone`, `/api/phone/routertest`, `/api/phone/modules`

Outside connections

- **DVSM** finds your public (WAN) address from `api.ipify.org`, falling back to `checkip.amazonaws.com`, with a 2-second limit each. Without internet the WAN rows say "Could not detect".
- **M17 host file** comes from `hostfiles.refcheck.radio`, falling back to `m17-project.github.io`. A download is only kept if it is at least 2 KB and lists at least 10 reflectors. The background poll checks every hour and fetches a new copy when the file is over 24 hours old.

Secrets

- The DVSM and STFU cards send the account passwords to your logged-in page, masked until you press the eye, so you can copy them into an app.
- The ASL-DVS checks report only whether a password is set, never its value. `var.txt` password lines are skipped completely.

Writes

- File saves go to `POST /api/asterisk/file`, `/api/dvswitch/file` and `/api/allmon3/file`. Each writes a temporary file, keeps the old file's owner and mode, syncs it to disk, and swaps it in place. No backup is kept. Asterisk file names are validated, and the DVSwitch and Allmon3 editors only accept the files in their fixed lists.
- Autopatch module changes edit only SysMon's own marked block in `modules.conf`, between `; >>> sysmon-phone modules` and `; <<< sysmon-phone <<<`. A conflicting `noload` elsewhere is commented out with `; sysmon-phone off:`. If the Dashboard's own block says noload, SysMon warns instead and leaves it, because Save Phone on the Dashboard would undo the change. Backups are named `modules.conf.sysmon-bak-YYYYMMDD-HHMMSS`, keeping 5. Live start and stop use `module load` / `module unload` through the Asterisk CLI.
- Restarts use `systemctl restart` on `stfu`, `usrp2m17` or `ircddbgatewayd`.

Defaults the checks treat as "not set"

Callsigns `N0CALL`, `NOTSET`, `NONE` or blank; the stock placeholder DMR and repeater IDs (such as `123456789`); the stock STFU server; area code `999`; and M17 address `0.0.0.0`.

System care

The last five SysMon tabs look after the hardware and the Pi itself: Tune (radio audio levels), Hardware (AMBE dongle, audio, power, video), SD Card, Security and Edit (SysMon's own settings). These tabs are where SysMon changes things most, so each section says what it writes and whether it restarts anything.

5.1 Tune tab

For a node with a radio on a USB sound interface run by Asterisk's **SimpleUSB** driver. A bridge-only node with no radio can ignore this tab.

SimpleUSB — Tune Settings

CONTROL	WHAT IT SETS
USB Device	The interface SimpleUSB is using (read-only).
RX Mixer Set	Receive level from the radio into the node, 0 to 1000. Slider or exact number.
TX Mix A Set / TX Mix B Set	Transmit level to the radio on each output channel, 0 to 1000.
Advanced settings	Carrier From and CTCSS From (where COS and tone detect come from: `no`, `usb`, `usbinvert`, `pp`, `ppinvert`); RX Boost; De-emphasis; Pre-emphasis; PL Filter; Invert PTT; Legacy Audio Scaling.
GPIO pins	Extra GPIO lines on the USB interface: in, out0 or out1. Pin 3 is PTT, and pins 5 to 8 exist on CM119 chips only.
Parallel port pins	Folded away: pins 1 to 17 as in, out, PTT, COR or CTCSS.
RX On Delay / TX Off Delay	Tick to turn on, then 0 to 100 frames.

An amber ● **unsaved changes** appears as soon as you move anything. **Reset** puts the card back to what is in the file. **Save** opens a dialog:

- **Save as preset**: also store the three levels as Radio 1 to 5, with a name such as "Mobile" or "Base".
- **Restart Asterisk after saving**: apply now. Without it, the file is saved but Asterisk keeps the old levels until it next restarts.

The Radio 1 to 5 buttons

The Radio buttons in SysMon's command bar (Chapter 3) are your saved presets. Hover over one to see its three levels. A `*` after the name means the levels have been changed since you applied that preset. Empty presets are greyed out, and so is every button when no SimpleUSB driver is loaded.

> **CAREFUL** Pressing a Radio preset button writes its levels to `simpleusb.conf` and **restarts Asterisk at once, with no confirmation**. Every link drops and the node is off the air for a few seconds. The button spins until Asterisk is back. Pressing the preset that is already applied does nothing.

Node Settings — rpt.conf

Folded away by default; press **Show**. It lists every node in `rpt.conf` with its role (radio, USRP bridge, hub, phone node) and four settings:

SETTING	RANGE	MEANING
duplex	0 to 4	How the node handles transmit and receive. 0 is right for a bridge node, 2 for a typical full-duplex repeater.
telemdefault	0 to 2	The default telemetry (voice announcement) mode: 0 off, 1 on, 2 the reduced setting. Bridge nodes use 0.
hangtime	0 to 60000 ms	How long the transmitter stays keyed after a signal drops.
linktolink	yes / no	Link-to-link radio mode.

Each value says whether it is *set on node*, inherited *from* a template, or the *default*. Nodes written by the Dashboard's phone setup, or set in another file, are locked here and say where to change them. Saving changes only the lines that differ, and can reload `app_rpt` so the change takes effect without a full restart.

5.2 Hardware tab

Copy hardware report at the top copies the whole tab as text, ready for a help request.

CARD	WHAT IT SHOWS AND DOES
AMBE Device	A ThumbDV or similar AMBE vocoder dongle, if present. You can **ping** the chip (it answers with its product ID), set the serial port speed (from a fixed list), add or remove a fixed `/dev` name for it with a udev rule, and **reset** it. Reset stops the DVSwitch services, resets the chip, then starts Analog_Bridge, MMDVM_Bridge and md380-emu first and the rest 3 seconds later.
Audio Device	The sound interface the node uses.
Power & Thermal	Supply health from the firmware: under-voltage, throttling and temperature limits, now and since boot.
Video & GPU Memory	Two switches that free memory on a headless Pi (below).
USB Bus	Every USB device, as a **Flat** list or a **Tree**. "possible AMBE" is only a guess from the chip maker. Only a badge with the exact product name, such as ThumbDV, is a confirmed match.
asl-find-sound	ASL's own sound-device finder, for SimpleUSB setup.
ALSA Cards	Linux sound cards, **Playback** or **Capture** side.

Video & GPU Memory

A Pi Zero 2 W used as a node never drives a screen, but the video driver and GPU still hold memory.

- **Disable HDMI and video driver** comments out the `dtoverlay=vc4-kms-v3d` line in `/boot/firmware/config.txt`.
- **Lower GPU memory** sets the GPU's share to 16 MB.

Both take effect after a **reboot**; until then the card marks them as pending. Each can be turned back on the same way.

> **CAREFUL** With the video driver off, a monitor on the HDMI port stays blank, and HDMI sound disappears. If any config file uses a sound card by number, the numbers can shift when the HDMI sound card goes away. When that is a risk, the card lists those lines so you can check them before you reboot.

> **NOTE** These two settings survive an instmon update. SysMon saves them when it is uninstalled for the update, and puts them back if it is reinstalled within 30 minutes (Chapter 2).

5.3 SD Card tab

A Pi runs from its SD card, and SD cards wear out. This tab tells you how the card is doing.

SD Card Health

SECTION	SHOWS
Warnings	Any problem found, in red at the top. The header reads "healthy" or the number of issues.
Device / Identity	The disk, maker, product, serial, manufacture date, firmware and hardware revision, erase size. On eMMC storage, also its own life-time estimate.
Capacity	Card size, and filesystem used and free. Amber from 75 %, red from 90 %. Also inodes used.
Filesystem (ext4)	State (should be `clean`), mount count, last check, lifetime writes, what happens on errors.
I/O counters	Bytes read and written since boot. This is a guide to wear, not a lifetime total.
Errors	Whether the root filesystem has dropped to **READ-ONLY** (usually corruption), and how many storage errors the kernel has logged, with the last lines.

Integrity Tests

Run fsck -n (advisory) checks the filesystem without changing anything, with a progress bar and a **Cancel** button. The result is only advisory because the card is in use while it runs. If you leave the tab, the test keeps running in the background.

> **TIP** To make the card last longer, use 44helper's System Optimization tab (Chapter 10). To keep a spare, take an image with instmon's Disk Image section (Chapter 13).

5.4 Security tab

14 passive checks of the node's security. The bar at the top counts Pass, Warn and Fail, with **Re-run checks** and **Copy**. The checks are grouped in four cards:

CARD	CHECK	SEVERITY
DVSwitch	BrandMeister hotspot password set	medium
AllStarLink	AMI port not reachable from the network	high
	AMI has no default or weak password	high

	IAX2 secret strength	medium
	Legacy `chan_sip` module disabled	low
	`/etc/asterisk` permissions	low
usrp2m17	USRP bridge ports bound to 127.0.0.1 only	high
	M17 gateway port exposure	high
Cross-cutting	SSH hardening	medium
	fail2ban jails cover the exposed services	medium
	Monit has authentication	medium
	Dashboard companion app requires a login	high
	Services run as non-root where possible	medium
	Service files and programs not writable by others	high

Click a check to open its explanation: **what it reads**, **what the result means**, **why it matters**, **how to fix it** and references. The amber Owner and Perms values on the Overview jump straight to the matching explanation here.

> **NOTE** The checks only read config files, `/sys`, `/proc`, `ss` and systemd. They never connect to AMI, M17, USRP or the Dashboard, so running them can't trigger fail2ban or fill a security log. Checks that read passwords report only yes or no.

> **CAREFUL** Take a FAIL on the USRP port check seriously. USRP has no password: anyone who can reach that port can key your transmitter under your callsign. Bind it to `127.0.0.1`.

5.5 Edit tab

SysMon's own settings. Change the fields, then press **Save** in the bar at the bottom of the page. **Reload from file** discards your changes.

SECTION	WHAT IT HOLDS
Visible Tabs	Tick the tabs to show. Overview and Edit are always on. A hidden tab stops polling, though an SD card test already running carries on.
Identity	Callsign, node and label for SysMon's own use.
Server	SysMon's port (9999) and bind address (0.0.0.0, meaning every interface). A change needs a SysMon restart, and a banner says so.
Thresholds	**NRestart warn** (3) and **NRestart crit** (10) colour the restart badge in the service panel. **CPU warn %** (50) and **RSS warn MiB** (200) turn the CPU and memory badges amber, with a ⚠, when a service is at or over them. All four load when the page opens.
Config Files	Folded away. Every Asterisk, DVSwitch and Allmon3 file with **Edit**, and **Hide** / **Show** to keep files you never touch off the ASL-DVS tab.
Pinned Services	Folded away. The Overview list as text: `unit | description | config_path | port | proto`, groups as `__GROUP__:Name`.

The bar also has four tools:

- **ASL AllConf** shows every Asterisk and Allmon3 file in one window, each under a header, with Copy All. It is ideal for sending your whole setup to someone helping you. **DVS AllConf** does the same for the DVSwitch files.
- **asl_dvs.conf** and **sysmon.conf** open the Dashboard's and SysMon's own config files in the editor.

> **CAREFUL** AllConf output includes passwords and secrets from your files. Remove them before you post it anywhere public.

5.6 Common tasks

Set and save radio levels

1. Tune tab: move RX Mixer Set while someone talks to you, until their audio sounds right on Allmon3 or a linked node.
2. Set TX Mix A (and B if used) for correct deviation on your radio.
3. Save, pick a preset slot and name, and tick Restart Asterisk.
4. From then on, one press of that Radio button brings these levels back.

Free memory on a Pi Zero 2 W

1. Hardware tab, Video & GPU Memory: turn on both options. Check any sound-card lines it lists.
2. Reboot from the command bar. Afterwards the card should no longer show "pending".

Is my SD card failing?

1. SD Card tab: look for red warnings, READ-ONLY, or kernel storage errors.
2. Run fsck -n. Errors in the result, along with kernel errors, mean it's time to image the card and replace it.

UNDER THE HOOD

What each tab writes

ACTION	FILE OR COMMAND	BACKUP
Tune Save	`/etc/asterisk/simpleusb.conf`; values range-checked; levels mirrored to `[radio_tune]` in `sysmon.conf`	none
Save as preset	`[radio_presets] radioN_values / radioN_title` in `sysmon.conf`	—
Restart Asterisk / preset apply	`asterisk -rx "module unload chan_simpleusb.so"` (allowed to fail), 2 s pause, `systemctl restart asterisk` (30 s limit). One restart at a time; the page polls `/api/radio/stack_status` until `core show uptime` answers.	—
Node Settings Save	Only changed keys inside the node's own `[node]` section in `rpt.conf`, comments kept; optional `module reload app_rpt`	none
Video / GPU memory	`/boot/firmware/config.txt`: tags the vc4 overlay line with `#ASL-DVS-VIDEO-OFF#`; adds `[all] gpu_mem=16` between `#ASL-DVS-GPUMEM-BEGIN` and `-END`	`config.txt.asl_dvs.bak` on every write
AMBE udev name	`/etc/udev/rules.d/99-ambe-sysmon.rules` (USB IDs: 4 hex digits; name: letters, digits, `_` and `-`, up to 32)	—
Edit Save	`/etc/sysmon/sysmon.conf` (identity, server, thresholds, pinned list, visible tabs, hidden files)	—

Reads

- **Power:** `vcgencmd get_throttled`. **Video:** `/proc/modules` (vc4), `/sys/class/drm/card*-HDMI-*/status`, `CmaTotal` in `/proc/meminfo`, `vcgencmd get_mem gpu` and `/proc/asound/cards`. A setting is "pending" when `config.txt` and the live state disagree.
- **SD card:** the root device's identity from `/sys/block`, `df`, `tune2fs -l` (needs root), the kernel I/O counters, the root mount options, and storage lines from the kernel log. The test runs `fsck -n` in the background; the page polls it every 1.5 s.
- **Security:** each check lists its exact source in its explanation. The same checks run from the shell with `sysmon.py --selftest`.

Boot-option carry-over on update

`--uninstall` records which boot options were on in `/etc/sysmon/boot_restore.json` (mode 600) and restores the stock `config.txt`. `--install` re-applies them if that record is under 30 minutes old, then deletes it. When updating from a SysMon too old to write the record, a fresh install falls back to `config.txt.asl_dvs.bak`, but only if that backup is under 30 minutes old and the service file is gone.

Full-build tabs

The full SysMon build, for a Pi 3, 4 or 5, adds five tabs the Pi Zero 2 W build leaves out to save memory: Firewall, Reg, Net, Zello and Console. Everything in Chapters 3 to 5 also applies to the full build, which has the same tabs.

FULL BUILD This chapter covers `sysmon_v6_13_81_20261013.py` (version `6.13.81`). On a Pi Zero 2 W you can install the full build with instmon's **Full build (override)** choice, but it uses about twice the memory.

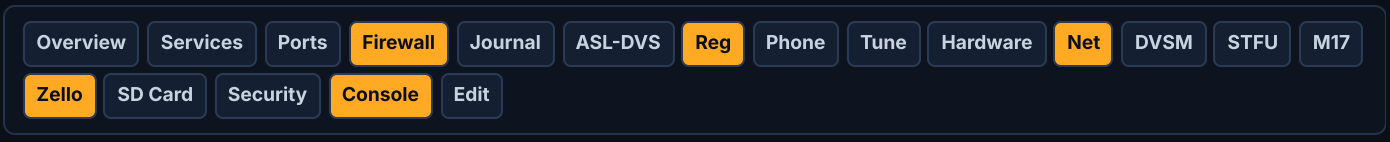


Figure 6.1 · Illustration. The full build's 19 tabs. The five highlighted are covered here.

6.1 Firewall tab

Shows and edits the Pi's own firewall. The **Backend** bar says which firewall SysMon found (ufw, firewalld, nftables or iptables) and whether it is active. A **Guide** explains ufw, firewalld zones and ASL3's default posture in plain words.

PART	WHAT IT DOES
Rule table	Number, port, action (ALLOW, DENY, LIMIT), source and the known service for each rule. Click a rule for details and **Delete**, which asks first. Deleting a firewalld service warns you if it is SSH or Cockpit.
Add Rule	Port, protocol (tcp, udp, any), action and an optional source address, then **+ Add**.
Zone details	For firewalld: which zone each interface is in. A port only opens if it is allowed in the zone of the interface traffic arrives on.
Raw firewall output	The full listing, including rules the table can't show (rich rules), with Copy.
Enable / Disable ufw	Turn the whole ufw firewall on or off. Both ask first.

CAREFUL **Enable ufw blocks everything that has no allow rule.** It asks first, and if ufw has no rule for the SSH port (from `sshd_config`, usually 22/tcp) it adds one before turning ufw on, so SSH stays open. It adds nothing else. Add ALLOW rules for the web pages (8989, 9999, 8990, 8991, 9997) from your network *first*, and IAX2 (4569/udp) so AllStar links keep working. SysMon before 6.13.75 enabled at once, with no prompt and no SSH rule.

On firewalld, SysMon adds each rule both live and permanently, so it survives a reload or reboot. 44helper's Firewall tab (Chapter 7) has a fuller zone-based editor with built-in guardrails, and works on both builds.

6.2 Reg tab

Checks how your node registers with AllStarLink, which is how other nodes find it.

- **HTTP Registration** reads `rpt_http_registrations.conf`; **IAX Registration** reads the `register =>` lines in `iax.conf`. For each: node numbers valid, a password present, the line syntax correct, and the live state from Asterisk (registered or not).
- **Dual registration** warns when both HTTP and IAX are active. Pick one and comment out the other; HTTP is ASL3's default.
- **Re-check** reads it all again. Live state also refreshes in the background every 30 seconds.

6.3 Net tab

CARD	WHAT IT DOES
Hostname & name announcer	Three steps, each with its command shown, **Run** and **Copy**: (1) set the hostname (letters, numbers and dashes, up to 63 characters); (2) install and start `avahi-daemon`, so the node answers to `name.local`; (3) test it. The IP address keeps working. You log in once more for each new name, and new backup and image file names use it. AllStar and DVSwitch settings aren't touched, and no reboot is needed.
WiFi	The current WiFi connection and signal.
Install mtr	Shown only if `mtr` is missing: installs `mtr-tiny`.
mtr trace	A 5-cycle route trace to a choice of targets, showing where along the path packets are lost.
Ping	5 pings to Google (8.8.8.8), allstarlink.org, or your DVSwitch servers.
Reflectors	Ping or trace any reflector configured on the Dashboard.

Only one job runs at a time; **Stop** ends it. Output appears in the card's console box.

6.4 Zello tab

For `asl-zello-bridge`, which links private node 1918 to a Zello channel. It is laid out like the STFU and M17 tabs (Chapter 4):

CARD	SHOWS
Installation	The bridge program (`/opt/asl-zello-bridge`) and its service.
Live Status	Read from the journal, not just whether the process is up: who keyed and unkeyed, and when. **Full Journal** opens the service panel.
Configuration	The settings in the service's systemd override (Free mode): Zello account and channel, USRP addresses, ports and gain.
Compatibility Checks	USRP ports 34012 / 32012, a `[1918]` node in `rpt.conf` that matches them, and credentials present.
Sample Override / Edit Override	A reference copy with placeholders, and the real file with **Start**, **Stop**, **Restart**, **Save** and **Copy**.

> **CAREFUL** The override holds your Zello password in plain text, and the editor shows it. Save writes it at once; Restart applies it.

6.5 Console tab

A live Asterisk console (`asterisk -rvvv`) in the browser, plus one-click buttons for the commands you'd otherwise type.

Switching it on and opening it

1. The console starts **switched off**. Press **Enable console** and enter the root password.
2. Press **Open console** and enter the root password again. You need it every time you open the console.
3. Type Asterisk CLI commands as usual. **Copy** and **Clear** work on the screen. **Close** ends the session.

- **Home network only.** It refuses any request that came through a tunnel or proxy (Cloudflare and the like), or from an address outside your LAN.
- **Never a shell.** The `!` key, which runs shell commands from the Asterisk CLI, is blocked, and so is Ctrl-Z.
- Only one console at a time. It closes after 10 minutes with no typing.
- **Disable console** switches it off again, and needs no password.

Command buttons

Pick a node in the **Node** list, then press a button; the result shows below, often with a plain-words summary.

GROUP	BUTTONS
Nodes	rpt stats, rpt lstats, node variables, link details, local nodes, registrations, **Disconnect all links**†
Phone	SIP registrations, endpoints, contacts, **Retry sign-in**†, SIP log for provider, transports, all calls, call quality, dialplan check, phone settings in Asterisk, phone nodes, phone node clash check, **Hang up all calls**†
AllStar links	IAX2 peers, registry, channels, link quality
Audio and CPU	codec conversions; audio (RTP) debug on/off; SIP logging on/off
Detail level	verbose 3, verbose 5, debug off
Reload	**Dialplan reload**†, **PJSIP reload**†
Memory	Swapping now?, programs in swap
System	uptime, version, modules check, **Restart Asterisk when idle**†

† Buttons that change something ask you to confirm. Debug and logging switch themselves off again after 5 minutes. The **Troubleshooting** card below the buttons runs read-only checks and explains the results.

UNDER THE HOOD

How the full-build tabs work

Firewall

- Backend detection order: ufw, firewalld, then nftables or iptables (raw listing only). ufw rules come from `ufw status numbered`; each row's number is what `ufw delete` removes.
- Add runs `ufw allow|deny|limit <port>/<proto>`, with `from <addr>` if given. On firewalld, SysMon runs `firewall-cmd --add-port` both with and without `--permanent`.
- Enable reads the SSH port from `sshd_config` (22 if unset), looks in `ufw show added` for an allow or limit rule on it, runs `ufw allow <port>/tcp` if there is none (and stops if that fails), then runs `ufw --force enable`. Disable runs `ufw disable`. Endpoint: `POST /api/firewall`.

Reg

Live state: `asterisk -rx "rpt show registrations"`, part of the background poll (every 30 s), with node numbers matched against the config files. Files: `rpt_http_registrations.conf`, `iax.conf`, `rpt.conf` and Allmon3's `allmon3.ini`.

Net

The hostname steps run `hostnamectl set-hostname` and update the `127.0.1.1` line in `/etc/hosts`, then install and enable `avahi-daemon` and `libnss-mdns`, and test the `.local` name from the node itself. Every command is shown before it runs, and Copy gives it to you for SSH if SysMon isn't running as root.

Zello

Service `asl-zello-bridge`; settings in `/etc/systemd/system/asl-zello-bridge.service.d/override.conf` as `Environment=` lines (`USRP_BIND`, `USRP_HOST`, `USRP_RXPORT`, `USRP_TXPORT`, gains and the Zello keys). Expected ports: RX 34012, TX 32012, node 1918. Live status parses `Keyed:` and `UnKeyed:` lines from the journal.

Console

- The on/off switch is the file `/etc/sysmon/console.enable`. It must be a plain file owned by root and not writable by anyone else, or the console stays off. The page writes who switched it on, from where and when. You can also create or remove it from a shell.
- Opening gives a one-time ticket, valid 60 seconds and tied to your login. The console then runs in a real terminal over a WebSocket, with its home directory in `/var/lib/sysmon/console-home`.
- Refused when the request carries proxy headers (`Cf-Connecting-Ip`, `Cf-Ray`, `X-Forwarded-For` and similar), or comes from an address that isn't loopback, link-local or private.
- The command buttons run fixed commands from an allow-list (`/api/console/run`). They don't need the console switched on or open. The node number comes from the list of your local nodes.

PART III

44helper

44helper is the node's setup and repair helper. It installs and removes the radio software, sets up remote access over 44Net or Cloudflare, runs system updates safely, trims SD-card wear, and saves and restores your configuration. SysMon tells you what is wrong; 44helper is usually where you fix it.

7. 44helper basics
8. 44Net and remote access
9. Software installers
10. Keeping the system healthy
11. The Config tab

44helper basics

How 44helper is laid out, how its step cards work, and the five status tabs: Overview, Services, Ports, Firewall and Actions Log. Open it at `http://<pi-address>:9997`. It is the same on every Pi.

7.1 The page

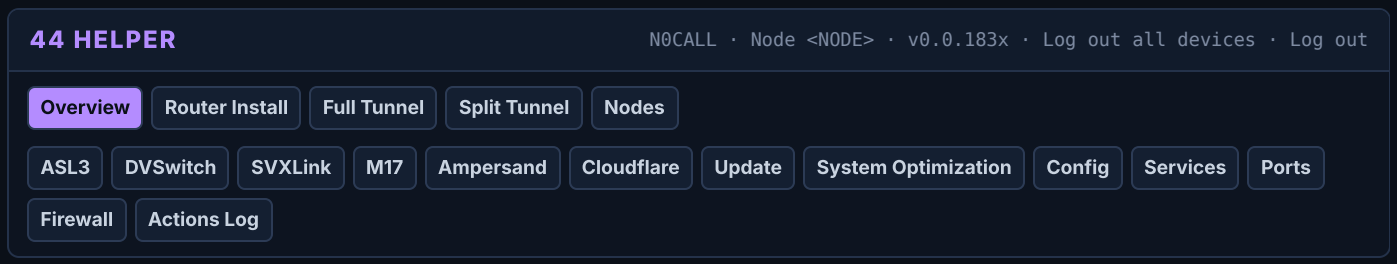


Figure 7.1 · Illustration. 44helper's header and tabs. Row one is 44Net remote access; row two is installers, maintenance, Config and the status tabs.

TAB GROUP	TABS	CHAPTER
Status	Overview, Services, Ports, Firewall, Actions Log	7
Remote access	Router Install, Full Tunnel, Split Tunnel, Nodes, Cloudflare	8
Installers	ASL3, DVSwitch, SVXLink, M17, Ampersand	9
Maintenance	Update, System Optimization	10
Configuration	Config	11

Logging in and out

- Log in with the root password. 44helper has its own login, separate from the Dashboard and SysMon.
- **Your login doesn't time out**, because install jobs can run for a long time. It ends when you press **Log out**, when anyone presses **Log out all devices** (which ends every 44helper login, on every browser), or when the Pi reboots.
- If 44helper restarts while you are logged in, the page reconnects to any running job by itself.

> **CAREFUL** 44helper runs as root and can install, remove and reconfigure anything on the Pi. Keep its port (9997) on your own network. Its Overview warns you if it ever becomes reachable through the 44Net firewall zone.

7.2 How step cards work

Most 44helper tabs are a list of numbered **step cards**. Each card does one job and shows exactly what it will run:

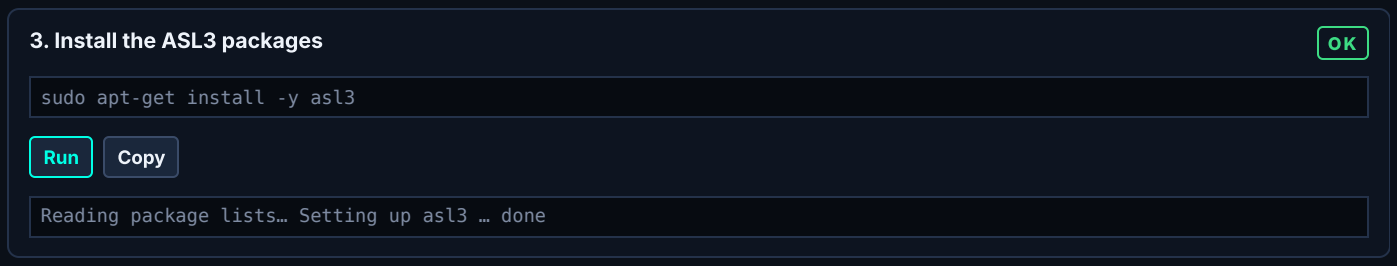


Figure 7.2 · Illustration. A step card: title and status, the command, Run and Copy, and the output.

PART	MEANING
Status pill	**Not run**, **Running…** (with elapsed time), **OK** or **Failed**. **Manual** means you run it yourself over SSH; **Config** means it saves a restore point.
Command box	The exact command. You can edit it before pressing Run, for example to change a version number. Manual steps are read-only.
Run	Runs the command on the Pi, as root, and streams the output below.
Copy	Copies the command (and its output) so you can run it over SSH or send it to someone.
▼ note	A folded note at the top of each tab, and on some cards. Open it before a first run: it says what the step changes and what can go wrong.

- Run the steps **in order**. Each one assumes the ones before it worked.
- Every install tab has a **Script** list (for example Bookworm or Trixie) and a **Custom** choice for typing any command.
- **Purge previous install** shows the removal steps. After a double confirmation they run one after another and stop at the first failure. Destructive steps, such as removing `/etc/asterisk`, are optional and only run if you press their own Run button.
- Step results last only while the page is open; a reload starts the cards fresh. A job that is still running carries on, and the card picks it up again.

> **CAREFUL** Whatever is in the command box runs as root, edits included. Only change a command when you know what the change does.

Safety checks before an install step

- **Masked services.** At the top of each install tab (and the Update tab) a card checks whether any service the tab needs is *masked* (blocked from starting). 44helper clears masks by itself and leaves those services **off**. You can **Turn on** each one, or **Put the mask back (on purpose)** so 44helper leaves it alone. While a mask can't be cleared, the card turns red and the tab's install buttons are locked.
- **Broken packages.** Before any step that installs or removes packages, 44helper asks the package system for unfinished or broken packages. If there are any, the step is refused with "NOT RUN" and a pointer to the Update tab's Package recovery steps (Chapter 10).
- **What to do next.** When an install step fails, a box opens under it with numbered repair steps. Each one has a button that jumps to the right Update-tab card, and a link brings you back.
- **Repair, then purge.** If a purge is refused because packages are broken, it offers to repair them in quiet mode and then run the purge again by itself.

The built-in terminal

A few steps need you to answer questions on screen, for example `asl-menu`, the USRP2M17 installer and Ampersand's first run. Those cards can open a real terminal in the page, but only:

- after the terminal has been **switched on** (from the page with the root password, or by creating `/etc/44helper/terminal.enable` over SSH);
- from your local network, never through a tunnel or Cloudflare;
- with the root password entered again each time you open it;
- for the one command printed on that card, nothing else.

One terminal at a time. It closes after 10 minutes with no typing, or when you log out.

7.3 Overview tab

The banner at the top sums up the worst finding. The cards below it:

CARD	SHOWS
Installed software	ASL3, DVSwitch, USRP2M17, SVXLink and Ampersand. Each is **installed** (with version and running services), **broken** (half-installed packages, or a package present without its program files), **leftovers** (removed but with settings, folders or unit files left behind), or **absent**. Click a name to go to its tab.
Tunnel / wg0	Whether the WireGuard interface `wg0` is up, its address and peer, and the last handshake.
Configured tunnels	The 44Net tunnels 44helper has set up (Pi-hosted or router).
44NetConnect firewall zone	The services and ports open to the 44Net side.
Self-check findings	Red if 44helper's own page, SysMon, the Dashboard, Cockpit or Asterisk AMI is open in the 44NetConnect zone. Amber if `wg0.conf` (which holds your private key) is not mode 600.

Re-check reads everything again straight away; otherwise the installed-software list is cached for 20 seconds.

7.4 Services tab

A grouped list of what is running: System (firewalld, systemd-resolved), 44Net Tunnel, AllStarLink / DVSwitch (Asterisk, the bridges, gateway command ports), M17, SVXLink, Ampersand, Cloudflare Tunnel, and **Dashboards (should stay private)**. Each line is a service or a listening port. Masked services are marked.

7.5 Ports tab

A reference table of every port the node's software uses, checked against what is actually reachable through your remote-access setup. The line above the table says where it looked: the Pi's 44NetConnect zone, the router's forward table, or "none (no tunnel configured yet)". **Running** means the port is listening on the Pi (or, for the outbound-only Cloudflare row, that `cloudflared` is up). From 44helper 0.0.179; before, most rows always read not running.

VERDICT	MEANS
OK	As it should be.

over-exposed	A port that must never be public is open: a dashboard, Cockpit, AMI, a loopback-only bridge port, or a gateway command port.
needed-but-not-open	A port other nodes must reach (IAX2 4569, EchoLink 5198-5199, VOTER 1667, Ampersand hub 4568) is running but not open, so inbound links will fail. Not shown until a tunnel is set up.
not-listening	Nothing is running on it, which is normal for software you don't use.

7.6 Firewall tab

44helper uses **firewalld**. The guardrail at the top warns when `wg0` is attached to the ASL3 image's own `allstarlink` zone instead of 44helper's `44NetConnect` zone.

CARD	WHAT IT DOES
Pi: 44NetConnect zone	What the 44Net side can reach. Add or remove the services `iax2`, `echolink` and `rtcm`, or a port such as `14569/udp`.
Quick port control (any zone)	Pick any zone (the active and default zones are marked) and **Show zone config**. Presets fill in IAX2, EchoLink, VOTER/RTCM or your DVSwitch Mobile port. Then: **Add (runtime test)** opens it until the next reload, a safe way to try it; **Make permanent** keeps that one port; **Remove** closes it everywhere; **Commit all runtime changes** keeps every test at once; **Reload firewalld**. An advanced section adds services and moves interfaces between zones.
Panic mode	If firewalld's panic mode (all traffic dropped) is on, a button turns it off.
Router: live port-forward table	With router access set up (Chapter 8), the router's forwards, each with a remove button.

As you type a port, a guardrail line checks it: a loopback-only port is **blocked** in every zone, and a dashboard, AMI or Cockpit port is allowed but flagged red.

> **TIP** Use **Add (runtime test)** first. If you lose access, a firewalld reload or a reboot puts things back. Make it permanent once you know it works.

7.7 Actions Log tab

- **Session log**: every command 44helper ran, with OK or FAIL, since it last started. **Refresh** reads it again.
- **Known-good router notes**: saved notes about router setups that worked. Click one to read it.

7.8 Common tasks

Check the node is safe before going on 44Net

1. Overview: the Self-check findings card should show no red lines.
2. Ports: no row should say **over-exposed**. IAX2 should be OK, not needed-but-not-open.
3. Firewall: the guardrail should be quiet, and the 44NetConnect zone should list only what you mean to open.

An install step says "NOT RUN"

1. Read the reason in its output: masked services, or broken packages.

2. For masks, use the card at the top of the tab. For broken packages, follow the "What to do next" buttons to the Update tab.
3. Come back with the link at the top of the Update tab and run the step again.

UNDER THE HOOD

How 44helper runs things

Files

PATH	HOLDS
`/usr/local/bin/44helper.py`	the program (service `44helper`, port 9997), a link into `/usr/local/lib/asl_dvs/44helper/`. Before 0.0.180 the program was `/opt/44helper/asl_dvs_m17_44helper.py`, which is now a second link to the same file for older instmon and uninstaller versions.
`/etc/44helper/44helper.conf`	settings: server, identity, tunnels, router access
`/etc/44helper/44helper_guard_keep.json`	services you chose to keep masked
`/etc/44helper/terminal.enable`	terminal switch: must be owned by root and not writable by others
`/run/44helper/sessions.json`	logins (mode 600, on tmpfs, so cleared at reboot)
`/etc/wireguard/wg0.conf`	the 44Net tunnel config (mode 600)

Jobs

- A step's Run posts `{mode, distro, step_id, command_text}` to its tab's `/api/<tab>/action`. The step must exist in that tab's script list; the command text itself runs through the shell, as root.
- **Package-changing commands** (apt, dpkg) run as their own systemd job, not inside 44helper. They keep going if 44helper restarts, and the page re-attaches to them. Package, build and install-script commands have no time limit; anything else stops after 15 minutes unless the step sets a different limit.
- Before a package command: an install step on a tab runs the masked-service check, which unmasks and disables any masked service the tab needs; then `dpkg --audit` must come back empty.
- The page polls `/api/action_status?job_key=…` about every half second, and keeps retrying if the connection drops.
- Every POST needs the login cookie and an `X-Requested-With` header (CSRF protection). Wrong passwords share one 5-try lockout across the login, the terminal and other password prompts.

Port roles

Each reference port has a role. **peer-inbound** ports (4569, 5198-5199, 1667 and 4568, all UDP) should be open when used. **mgmt-LAN** ports (5038, 8080, 8989, 9090, 9997 and 9999) and **loopback-only** ports (31100, 31103, 32008, 34008, 32012, 34012, and the command ports 6073-6075 and 54321) must never be exposed. **outbound-only** (7844, Cloudflare) never needs an opening. "Exposed" means in the 44NetConnect zone (Pi-hosted), in the router's forwards (router, single address), or the whole subnet (router, routed subnet with masquerade off).

44Net and remote access

How to make your node reachable from the internet, or reach it yourself while away. 44helper offers three routes: a 44Net tunnel run on the Pi (Full or Split), a 44Net tunnel run on your router, or a Cloudflare Tunnel. This chapter helps you pick one, then walks through each tab, and finishes with the Nodes tab for several nodes on one network.

8.1 Which route?

ROUTE	TAB	BEST FOR	WHAT YOU NEED
Pi tunnel, Full	Full Tunnel	A node that must be reachable by other AllStar nodes, with all its traffic going out through 44Net.	A 44Net Connect account and tunnel.
Pi tunnel, Split	Split Tunnel	The same, but only 44Net traffic uses the tunnel. DMR, M17 and everything else take your normal internet. **Usually the better choice.**	The same, plus one edited line.
Router tunnel	Router Install	Putting several devices, or a whole routed 44Net subnet, behind one tunnel on an OpenWrt / GL.iNet router.	SSH access to the router; a single 44Net address or a routed subnet.
Cloudflare Tunnel	Cloudflare	Reaching the web pages yourself from anywhere, with no port opened at home.	A free Cloudflare account and domain.

> **NOTE** 44Net (AMPRNet) gives amateur stations real public 44.x.x.x addresses. 44Net Connect is ARDC's free WireGuard service that hands you one. Other AllStar nodes can then reach your IAX2 port directly, with no port forwarding on your home router.

8.2 Full Tunnel and Split Tunnel tabs

Both tabs set up a WireGuard tunnel (`wg0`) on the Pi, with a firewalld zone called **44NetConnect** that decides what the 44Net side can reach. The only difference is the `AllowedIPs` line in the tunnel config:

TAB	ALLOWEDIPS	EFFECT
Full Tunnel	`0.0.0.0/0`	All the Pi's outbound traffic goes through 44Net. This is the config exactly as the 44Net portal gives it.
Split Tunnel	`44.0.0.0/9, 44.128.0.0/10`	Only 44Net addresses use the tunnel. You edit this line yourself before pasting.

The steps

1. **Firewall prereq**: installs and starts firewalld.
   1a. systemd-resolved prereq: installs WireGuard and systemd-resolved, without which the tunnel can fail to start. A reboot is recommended after this, before step 6.
2. **Create the 44NetConnect zone.**

3. **Add inbound services and ports**: tick what other stations may reach. AllStarLink IAX2 and EchoLink/SvxLink are ticked by default; VOTER/RTCM is optional. Asterisk AMI can be added but is flagged red. Full Tunnel also has fields for your DVSwitch Mobile port and a custom port. Split Tunnel offers only IAX2 and EchoLink, because nothing else needs to be reachable.
4. **Paste WireGuard config**: copy the whole `[Interface]` / `[Peer]` block from your tunnel on connect.44net.cloud, paste it, and press **Write wg0.conf (chmod 600)**. It refuses a config with no real private key. Leave **Also back up a copy to the instmon library** ticked, so you can recover the tunnel after re-flashing the card.
   4a. DNS & MTU: optional DNS handling (off, all DNS through the tunnel, only 44Net names, or a `DNS =` line) and an MTU value.
   4b (Split): **Repair AllowedIPs** fixes a config pasted unedited. **4b (Full)**: optional outbound rules for your digital-voice reflector ports.
5. **Attach wg0 to the zone.**
6. **Enable + start the tunnel**, then **Verify**: checks your public address through connect.44net.cloud/myip and Asterisk's AllStar registrations.

> CAREFUL Starting the tunnel can change how the Pi answers the network. If you are connected over SSH or Cockpit from outside your home network, that session may drop. Do step 6 from home, or on a keyboard and screen.

> CAREFUL A Full Tunnel config pasted into the Split Tunnel tab unedited is the classic cause of "AllStar works but DVSwitch and M17 don't": every DNS lookup goes into the tunnel. Use step 4b's Repair button, then restart the tunnel.

Other buttons

- **Troubleshooting: bridge startup (netcheck)**, below step 6: a read-only scan for when Analog_Bridge, MMDVM_Bridge or STFU won't start after the tunnel comes up. It is most likely on a Full Tunnel.
- **Convert to Split / Full Tunnel**: switches an installed tunnel to the other kind, with a preview first.
- **Purge**: undoes the tunnel and the zone.
- If one kind is installed and you open the other tab, its steps are locked until you press **Acknowledge and unlock**, so you can't overwrite a working tunnel by accident.

8.3 Router Install tab

For an OpenWrt-based router (GL.iNet models are recognised), driven over SSH with a key. Two models:

- **Model B, single address.** The router holds your one 44Net Connect address and forwards chosen ports to the Pi.
- **Model C, routed subnet.** A coordinator gives you a 44.x subnet; the router's LAN uses it, so the Pi and other devices get real 44Net addresses.

STEP	WHAT IT DOES
1. Router access	Access method (none, ssh, api), host, user, SSH key path (for example `/etc/44helper/router_id_ed25519`) and poll interval. **Save**, then **Test connection / Re-probe**.
2. Firmware/model detection	OpenWrt? `uci` available? GL.iNet model? Read-only.

2a. LuCI	Installs the OpenWrt web interface if it is missing.
3. Allocation type	Choose Model C (routed subnet) or Model B (single address). The steps below change with it.
4. Capture config	Paste the WireGuard config, plus the LAN subnet for Model C. The private key stays on the Pi; the browser never sees it again.
5. Apply to router	Pushes the config to the router with `uci`.
6. Set router LAN IP *(Model C)*	Moves the router's LAN onto the 44Net subnet. **Your connection to the router and the Pi may drop**: renew the Pi's address or reboot it, then update the host in step 1.
7. Bring up the tunnel	`ifup wgclient`, then checks its status.
8. Pi's address	Looks up the Pi's address in the router's DHCP leases. Model C can add a static binding.
9. Port forwards *(B)* / Fix firewall zone *(C)*	B: tick the services to forward from the router's 44.x address to the Pi. C: sets the WireGuard zone to forward, with masquerade off, so devices keep their real addresses.
10. Verify	B: checks outbound traffic uses the 44.x address. C: traces to portal.ampr.org. **Save known-good note** records the model, firmware and access method (no secrets) for the Actions Log.

> **NOTE** Model B's Verify can't prove that other stations can reach you from outside, because 44helper has no way to test inbound from the internet. Ask another node to link to you.

8.4 Nodes tab

For setups with more than one AllStar node on the same network, such as two Pis behind one router. Each box needs its own IAX2 port, and the nodes need to know about each other.

CARD	SHOWS
This Box	This Pi's IAX2 `bindport` and whether it falls in the recommended range (4560 to 4580). **Edit bindport** changes it.
Local Nodes	The nodes defined on this Pi, with their `rxchannel` and any findings. **Edit** opens just that node's line.
Peer Nodes	Other nodes on your network. **Wired** means this Pi's `rpt.conf [nodes]` already has a line for it. Add one with Node #, IP, port, label and, optionally, its 44helper address. A 🔗 **44helper** button then opens the peer's 44helper, dimmed if it isn't answering.

Each finding has a **What to do** line. Saving an edit shows the line first, backs the file up, then shows the change.

8.5 Cloudflare tab

A Cloudflare Tunnel is an outgoing connection from the Pi to Cloudflare. Cloudflare then serves a hostname you choose, such as `node.example.com`, and passes the traffic down that connection. Nothing is opened on your router.

token method (persistent)

1. **Pre-flight** (read-only): is cloudflared already installed, which Node Controller pages are running, and how it fits with any WireGuard tunnel.
2. **Install cloudflared** from Cloudflare's package repository.

3. **Install the connector service** (manual): copy the `sudo cloudflared service install eyJ…` command from your tunnel in the Cloudflare Zero Trust dashboard, and run it over SSH. It is done by hand on purpose, so your token never lands in 44helper's log.
4. **Verify the connector**: service status and recent log. The Zero Trust dashboard should show the connector as Healthy.
5. **Map a Public Hostname and add an Access policy** in the Zero Trust dashboard, for example `http://localhost:9999` for SysMon.

quick method (test only)

Run `cloudflared tunnel --url http://localhost:8989` over SSH. It prints a random `*.trycloudflare.com` address, needs no account, and stops when you press Ctrl-C.

> **CAREFUL** A Cloudflare Tunnel **bypasses the firewall**. Nothing opens inbound, so 44helper's Ports and Firewall tabs can't see what you have published. Anything you map is reachable by the whole internet, protected only by that page's own login. Always put a **Cloudflare Access policy** (for example, email one-time codes for your address only) in front of every hostname. Never publish 44helper itself.

> **NOTE** Cloudflare works with no WireGuard tunnel or with a Split Tunnel. Under a Full Tunnel it still works, but its connection rides the 44Net tunnel and drops whenever the tunnel does.

8.6 Common tasks

Put a node on 44Net with a Split Tunnel

1. On connect.44net.cloud, request a tunnel and wait for approval. Copy its config and change `AllowedIPs` to `44.0.0.0/9, 44.128.0.0/10`.
2. Split Tunnel tab: run steps 1, 1a and 2, reboot, then step 3 with IAX2 ticked.
3. Paste the config in step 4 and write it. Run step 5, then step 6 from home, then Verify.
4. Check the Overview's Self-check findings and the Ports tab (Chapter 7) before you rely on it.

Reach SysMon from anywhere

1. Cloudflare tab, token method: run steps 1 and 2, then step 3 over SSH.
2. In the Zero Trust dashboard, map `sysmon.yourdomain` to `http://localhost:9999`, and add an Access policy for your email address.
3. Open the hostname, pass the Access check, then log in to SysMon as usual.

UNDER THE HOOD

What gets written where

Pi tunnel

- `/etc/wireguard/wg0.conf`, written mode 600 through a temporary file. The page never shows the private key again; status shows "present, permissions 600". The library copy goes to `/etc/asl_dvs/instmon_library/config/wg0.conf`.
- The zone is `/etc/firewalld/zones/44NetConnect.xml`, managed with `firewall-cmd --permanent`. When firewalld's polkit crash ("unable to guess signature from an empty dict") stops a command, 44helper edits the zone XML directly and reloads.
- The tunnel runs as `wg-quick@wg0`, or a custom fallback unit `44net-tunnel.service` where that template isn't available.
- Which kind is installed is stored as `tunnel.pi.kind` in `[tunnels]` of `44helper.conf`; the DVSwitch Mobile port as `tunnel.pi.dvswitch_mobile_port`.
- The guardrail also checks the ASL3 image's own `allstarlink` zone: `wg0` must not be in it.

Router

- SSH with `BatchMode=yes`, a 6-second connect timeout and `StrictHostKeyChecking=accept-new`, using the key you name. Probes are cached for the poll interval (at least 5 s, default 60).
- Changes use `uci` on the router (firewall redirects for Model B, `firewall.wgzone` forward=ACCEPT and masq=0 for Model C, `network.lan` for the LAN address), then `/etc/init.d/firewall reload` or a network reload.
- The Ports tab reads forwards back with `uci show firewall`.

Nodes

Reads `iax.conf` (bindport), `rpt.conf`, `extensions.conf` and `allmon3.ini`, at the paths set in `[nodes]` of `44helper.conf`. Peers you add are stored there too (`peers`, `peer_links`). Peer 44helpers are probed for reachability and version.

Cloudflare

`cloudflared.service`, with its token and credentials in `/etc/cloudflared` (removed only by the optional purge step). The edge connection is outbound 7844/udp (QUIC), falling back to TCP. Only one cloudflared service is allowed per machine.

Software installers

Five tabs install, and remove, the radio software a node runs: ASL3, DVSwitch, SVXLink, M17 (USRP2M17) and Ampersand. All five use the step cards from Chapter 7: pick a script, run the steps in order, and read each card's note first.

9.1 Before any install

- **Pick the right script.** Bookworm is Debian 12; Trixie is Debian 13. Check yours with `cat /etc/os-release`.
- **Pause the node on a Pi Zero 2 W.** Turn on quiet mode on the Update tab (Chapter 10), or use instmon's Quiet System. A big install competes with everything else for the SD card.
- **Watch the masked-service card** at the top of the tab. Install buttons stay locked while it is red.
- **Afterwards**, check the Ports tab (Chapter 7) so nothing you just installed is exposed by mistake.

> **TIP** Several tabs end with **Save original configs as Restore point**. Run it once, after *all* the software you use is installed and set up. It saves every installed system's config files as your Restore point (Chapter 11). Running it again only adds systems installed since; saved files are never overwritten.

> **NOTE** Cockpit, the web admin console on port 9090, has install steps on several tabs. Install it once from any tab; the others skip it. Its purge step is optional and never runs automatically, because removing it affects the whole node.

9.2 ASL3 tab

Installs AllStarLink 3 from the official repo.allstarlink.org packages. Script: Bookworm or Trixie.

#	STEP
1–4	Prerequisites (wget, curl, git, firewalld, alsa-utils…); download and install the ASL3 repository package; refresh the package list.
5–6	Install ASL3 core (`asl3`) and Allmon3.
7–8	Install and enable Cockpit.
9	Node database and text-to-speech utilities.
10	*Optional:* full ASL3 Appliance branding and firewall defaults (`asl3-appliance`, `asl3-appliance-pi`, which pulls in apache2).
11–13	Remove the default Allmon3 account, set your own (fill in Username and Password, not the command), and enable Allmon3.
14	Enable the node-database update timer.
15	**Manual:** `sudo asl-menu` to set node number, callsign and radio interface. It is a full-screen menu: run it over SSH, or in the built-in terminal (Chapter 7).
16	Save original configs as Restore point.

Purge: stops the services, purges the ASL3 and Allmon3 packages, removes unused dependencies, the repository and the downloaded file. *Optional, manual only:* remove `/etc/asterisk` (your node settings), and Cockpit.

> **CAREFUL** The Allmon3 password step puts the password on the command line, so it is briefly visible in the process list. Don't reuse a password that matters elsewhere.

9.3 DVSwitch tab

Installs DVSwitch Server (Analog_Bridge, MMDVM_Bridge, md380-emu and the gateways) from dvswitch.org's official repository.

SCRIPT	STEPS
Bookworm	Download and run the repo-install script; refresh; install DVSwitch Server; enable the DVSwitch services and the digital-mode gateways at boot; set callsign, DMR ID and BrandMeister password.
Trixie	Starts with **Check what's already here** (read-only), then the same repository (DVSwitch has no Trixie packages yet), plus a **Test install (changes nothing)** step as a safety gate. **If the test install fails, stop.**

- **Set Callsign / DMR ID / BrandMeister password** fills in `MMDVM_Bridge.ini` from three fields, then restarts the bridges. The DMR ID takes your 7-digit ID plus a 2-digit suffix, for example `320123401`. Check the file afterwards if you use more than one DMR network.
- **Web-server clash:** ASL3's apache2 (Allmon3) and DVSwitch's lighttpd dashboard both want port 80. Step 1 reports a `CLASH`. Two optional steps fix it: **turn off the DVSwitch web dashboard**, or **move it to another port** (8081 by default; ports used by other suite tools are refused). Both do nothing if there is no clash.

Purge is the same for both scripts: it removes anything either could have installed. Optional steps remove leftover config and leftovers from an old third-party build script.

9.4 SVXLink tab

SCRIPT	WHAT IT INSTALLS
USRP fork (default)	Builds DL1HRC's unofficial SvxLink fork from source. It adds a UsrpLogic module, so SvxLink can talk to the DVSwitch bridges over USRP. Steps: conflict check, build dependencies, clone, configure and build, check the USRP part built, create the svxlink user, locate files, switch UsrpLogic on in the config, and a **hand-written** systemd unit (shown on its card for you to create). Installs under `/usr/local`.
Bookworm / Trixie	Debian's own `svxlink-server` package: install, ALSA utilities, add the svxlink user to the audio and serial groups, enable at boot, set callsign. Sound-card and PTT/COS settings stay at their defaults; set them by hand for your hardware.

> **CAREFUL** The USRP fork is unofficial and bench-tested only. It is not merged upstream, and its build-step note flags an unpatched crash from a crafted packet. The package and the fork can't run at the same time. One purge covers both: it stops both services, purges the package, and offers the fork's files as optional steps.

9.5 M17 tab

Installs **USRP2M17-Selector** (mjwgeek, GitHub): the USRP2M17 bridge plus a small web page for picking M17 reflectors. It runs the project's own unmodified installer. The project's licence allows personal, non-commercial

use only, with no modification or redistribution.

STEP	KIND
Install git, PHP for Apache; clone USRP2M17-Selector	Run
Run the installer (asks for your callsign)	**Manual**, or in the built-in terminal
`rpt.conf` **: add the M17 node stanza (node 1917)**	**Manual edit**: text you apply by hand
`modules.conf` **: enable** `chan_usrp`	**Manual edit**
`extensions.conf` **: add the M17 extension**	**Manual edit**
Restart Asterisk; install the sudoers rule for the web page	Run

The three Asterisk file edits are shown as text rather than run automatically, because editing a live Asterisk config unattended is too risky. **Purge** removes the service, web page, sudoers rule and unit, but **never** undoes those three edits; revert them by hand.

> **TIP** After installing, check SysMon's M17 tab (Chapter 4). It confirms the ports (32008 and 34008) and the `[1917]` node match.

9.6 Ampersand tab

Ampersand is a separate, Asterisk-free AllStar/IAX node project. Pick a **Product**:

PRODUCT	STEPS
Radio node (`amp-server` **)**	Conflict check; required packages; udev HID rule (COS/PTT access); download and unpack the server; **first run** in the foreground as a smoke test (built-in terminal); run as a systemd service.
Hub node (`amp-hub` **)**	Conflict check; wget; download, unpack and run its `install.sh`; configure `/usr/etc/amp-hub.env`; enable and start the hub.

> **CAREFUL** Ampersand **replaces** Asterisk; the two don't share a node. If this Pi already runs ASL3 on IAX 4569/udp, give Ampersand its own node number and IAX port (the hub defaults to 4568), or use a separate Pi. The tab never stops Asterisk or changes ports for you. The download date is pinned in the command box, so change it if a newer build exists.

9.7 Purge clean-up steps (every tab)

Every purge list ends with four extra steps that only touch that tab's own packages and services:

1. **Purge removed-but-not-purged packages.** Their leftover masks are what break the next install.
2. **Unmask this tab's own services.** Debian's intentional masks are never touched.
3. **Remove broken boot links** for this tab's services.
4. **Still left behind** (read-only): packages, masks, service files and folders that remain, so nothing is left silently.

9.8 Common tasks

Build a new bridge node from a fresh Raspberry Pi OS

1. ASL3 tab: run steps 1 to 14, then run `asl-menu` (step 15).
2. DVSwitch tab: run every step, including callsign and DMR ID. Fix a port 80 clash if step 1 reports one.
3. M17 tab, if you want M17: run the steps and apply the three file edits by hand.
4. ASL3 tab step 16: save the Restore point.
5. SysMon: work through the ASL-DVS, DVSM, STFU and M17 tabs until every FAIL is gone.

Start a broken install over

1. On the tab, press **Purge previous install** and confirm twice.
2. If it stops because packages are broken, press **Repair, then purge**.
3. Read the **Still left behind** report, then run the install steps again.

UNDER THE HOOD

How the installers are built

- Each tab's steps live in a Python table (`_ASL3_SCRIPTS`, `_DVSWITCH_SCRIPTS`, `_SVXLINK_SCRIPTS`, `_M17_SCRIPTS`, `_AMP_SCRIPTS`, `_CF_SCRIPTS`), keyed by install or purge and then by script. The page fetches them from `/api/<tab>/script`.
- Step flags: `manual` (copy only), `fields` + `cmd_template` (form inputs substituted into `{{name}}`), `config_save` (Restore point), `auto: False` (left out of the automatic purge pass), `confirm_text` (an extra prompt), and `timeout`.
- apt commands use `DEBIAN_FRONTEND=noninteractive` with `--force-confdef --force-confold`, so an upgrade keeps your edited config files instead of stopping to ask.
- Each step runs in its own shell, so a `cd` doesn't carry over; steps use absolute paths such as `/tmp/asl-apt-repos.deb13_all.deb`.
- The services each tab's masked-service check covers: ASL3, Asterisk, Allmon3, apache2 and the two ASL3 timers; DVSwitch, the bridges, md380-emu, Analog_Reflector, ircDDBGateway, netcheck, monit, lighttpd, mosquitto and stfu; SVXLink, svxlink, svxlink-usrp and remotetrx; M17, usrp2m17; Ampersand, amp-server and amp-hub; Cloudflare, cloudflared. The Update tab checks all of them.
- The lighttpd move step backs up `lighttpd.conf` once (`lighttpd.conf.44helper-bak`), and checks the new config with `lighttpd -tt` before restarting.
- The Overview decides installed, broken, leftovers or absent from `dpkg-query` status letters, key program files (for example `/opt/MMDVM_Bridge/MMDVM_Bridge`) and unit states.

Keeping the system healthy

Two tabs look after Linux itself. The **Update** tab updates packages safely, with quiet mode and a set of package-recovery steps for when an update goes wrong. **System Optimization** cuts down SD-card wear on a small Pi.

10.1 Update tab

Quiet mode

Before a big upgrade or install, quiet mode pauses everything that competes for the SD card, in this order:

1. watchdogs and update timers (the Node Controller watchdog, monit, the ASL3 node-list and database timers, man-db);
2. background disk users (the performance recorder, PackageKit, the manual-page indexer);
3. the suite web tools (SysMon, the Dashboard, instmon, and the DVSwitch web page);
4. the radio services (Asterisk, Analog_Bridge, MMDVM_Bridge, md380-emu, SvxLink, ircDDBGateway, STFU).

Turn off restarts exactly what was paused, in reverse order. 44helper, wifimon, SSH, Cockpit, networking and the firewall are never touched. The card previews the exact commands before you press anything. A banner across the top of every 44helper page reminds you while quiet mode is on.

> **CAREFUL** The node is **off the air** while quiet mode is on, and it stays on until you turn it off. Unlike instmon's Quiet System, it has no 30-minute auto-restore. You can't turn it on while a package job is running. If something fails to come back, the card shows the likely cause and offers **Reboot node**.

Package lookups and status

- **Package Lookup — Installed** lists every installed package. **Version Policy** shows installed and candidate versions for one package name, for example `dvswitch-server`.
- **Upgradable Packages** and **Held Packages** cards, and a banner saying whether a **reboot is required**. **Re-check** refreshes them.

The update steps

Numbered steps, several with alternatives shown underneath. Steps that change packages have a **Run in quiet mode** box; leave it ticked on a Pi Zero 2 W.

#	STEP	ALTERNATIVES
1	Refresh package index (`apt update`)	
2	Fix missing/broken packages (`dpkg --configure -a`, then `apt install -f`). Safe any time.	Is apt still running? (read-only) · Reconfigure half-installed packages
3	Upgrade packages (`apt upgrade`)	Full upgrade: can *remove* packages to satisfy new dependencies. Check the Upgradable list first.
4	Autoremove unused packages	Autoremove + purge configs (not reversible)
5	Clean local package cache	Autoclean (keeps recent downloads)

6	*Optional:* install Webmin (port 10000, HTTPS)	Remove Webmin
7	*Optional:* install Ajenti 2	Remove Ajenti 2
8	*Optional:* install fail2ban (SSH jail)	Check status (read-only) · Remove fail2ban
9	**Package recovery 1: check** (read-only)	
10	**Package recovery 2: finish half-installed packages**	
11	**Package recovery 3: re-install broken packages**	

CAREFUL **fail2ban:** before running step 8, edit the `ignoreip` line in its command box. It ships with `192.168.1.0/24`, which is probably not your network. Add your tunnel subnet too if you manage the node over 44Net. A ban lasts 10 minutes and covers SSH (port 22) only, so the web pages stay reachable. To lift a ban, use Custom with `fail2ban-client set sshd unbanip <your-ip>`.

When an install or upgrade was cut off

A power cut or crash in the middle of an upgrade can leave packages half-installed. 44helper then refuses further package steps until this is fixed (Chapter 7). Use the recovery steps:

1. **Step 2's "Is apt still running?"**: wait until it says nothing is running.
2. **Step 9 (check)**: lists any problems. If it lists any, **don't reboot yet**. A half-installed kernel or firmware package can stop the Pi from starting.
3. **Step 10**: finishes ordinary packages first, then kernel and firmware last, at low priority. It stops at the first failure.
4. **Step 11**, only if step 9 or 10 says something needs a reinstall: re-installs from the copy already on the Pi, or downloads just that package.
5. **Step 9 again**: it should list nothing. Now it is safe to reboot.

TIP A failed install step on another tab opens a **What to do next** box. Its buttons jump straight to these cards, and a link at the top of the Update tab takes you back.

10.2 System Optimization tab

Each card shows the current state, an action, a **Revert** that puts back exactly what was there before this tab changed it, and **Copy**. The exact commands are shown on every card.

CARD	WHAT IT DOES	TAKES EFFECT	UNDO
Root fstab — noatime	Stops the card recording a write every time a file is read.	at once (remount)	Revert; one backup of `/etc/fstab` is kept
journald — volatile	Keeps the system log in memory instead of on the card.	at once (journald restart)	Revert
rsyslog	Stops and disables the second, text-file logger.	at once	Revert
dphys-swapfile — Disable	Turns off the swap file on the card.	at once	Revert (works offline)

dphys-swapfile — Uninstall	Removes the swap-file package entirely, after showing what apt would remove.	at once	Reinstall, **needs internet**
mandb — Disable rebuild	Stops the periodic manual-page index rebuild.	at once	Revert
mandb — Remove package	Removes `man-db` entirely.	at once	Reinstall, **needs internet**

CAREFUL With journald volatile and rsyslog off, **no log survives a crash or power cut**. That is the point, but remember it before you go looking for why a node rebooted. Existing logs already on the card are not deleted.

CAREFUL Turning swap off on a 512 MB Pi Zero 2 W is a real out-of-memory risk under load. Check your memory headroom first (Chapter 2, and `free -m`). Many Pi Zero 2 W systems use zram (compressed swap in memory) instead; that is separate from dphys-swapfile and isn't touched here.

10.3 Common tasks

Monthly update on a Pi Zero 2 W

1. Config tab: take a save point (Chapter 11).
2. Update tab: turn quiet mode on.
3. Run steps 1, 2 and 3, all with Run in quiet mode ticked. Read the Upgradable list before choosing Full upgrade.
4. Run step 4, then step 9 to confirm nothing is half-installed.
5. Turn quiet mode off. If the banner says a reboot is required, reboot.

Make the SD card last

1. Apply noatime, journald volatile and rsyslog off. These are low risk and undoable.
2. Disable the mandb rebuild.
3. Leave swap alone unless you know the node has memory to spare.

UNDER THE HOOD

How updates and optimizations are applied

Quiet mode

- The state is kept in `/etc/44helper/44helper_quiet_state.json`: what was stopped, and why. That is how Turn off restarts exactly that set, even after a 44helper restart.
- Matching uses unit patterns per tier (for example `*sysmon*.service`). Anything matching the never-touch list (`*44helper*`, `*wifimon*`, `ssh*`, `cockpit*`, `NetworkManager*`, `firewalld*`, `systemd-*`, `dbus*`, `wpa_supplicant*`, `polkit*`) is skipped. A running `mandb` is killed between tiers.
- Turning on is refused while any dpkg or apt lock is held. Each stop waits up to 150 s.
- Reboot node schedules `systemctl reboot` 3 seconds later with `systemd-run`, so the page can show the reply first.

Update steps

- Package steps run as systemd jobs. Ticking **Run in quiet mode** wraps the job in quiet mode, at low CPU and disk priority. The recovery steps run under `nice -n 19 ionice -c2 -n7`.
- Step 10 finishes packages whose status letter is U, F, W or t. It does kernel and firmware packages (`linux-image`, `raspi-firmware`, `initramfs-tools` and similar) in a separate last pass, then runs `dpkg --configure --pending`.
- Step 11 reinstalls packages flagged R, or with missing list or md5sums files, from `/var/cache/apt/archives`. It falls back to `apt-get download` into `/tmp/44helper-debs`.
- fail2ban writes `/etc/fail2ban/jail.d/44helper-fail2ban.conf` with `backend = systemd`, which is needed once rsyslog is off and `/var/log/auth.log` no longer exists.

System Optimization

- noatime: the candidate fstab is checked with `findmnt --verify` before it is written, then applied with `mount -o remount,noatime /`.
- journald: a drop-in at `/etc/systemd/journald.conf.d/44helper-sysopt.conf`; Revert removes it.
- Each Revert puts back the state saved the first time that card changed something, not a fixed default. "Nothing to revert yet" means this tab hasn't changed it.

The Config tab

The Config tab is your safety net for the node's configuration. It saves copies of every config file, restores them in one go, carries node wiring from one Pi to another, compares your files with the stock ones, and checks that callsigns, IDs and ports agree everywhere. Use it before and after any change.

11.1 The ideas

TERM	MEANS
System	A group of files restored together: **ASL3**, **DVSwitch**, **USRP2M17** or **Phone**. Only installed systems are saved.
Restore point	The original config, saved **once** after your installs are finished. Your "known good" baseline.
Save point	A named copy of the current files, taken whenever you like, for example "before reflector change".
Automatic copy	Taken by itself before every restore, preset, standard-value change or undo, so every change on this tab can be undone.
Preset	Just the *wiring* of a node (private nodes, ports, extensions), with no identity, for copying to another Pi. Today there is one: **Travel Node**.

Up to **10** points are kept in all. When a new automatic copy would go over the limit, the oldest automatic copies are deleted first; your named save points are never removed for you.

11.2 Restore point

- **Save Restore point**: run it once, after ASL3, DVSwitch and USRP2M17 (whichever you use) are installed and working. The same button is the last step on the install tabs. Saving again only adds systems installed since; files already saved are never overwritten.
- The status lists each system and file. A file that has changed since the Restore point has **View diff**, which shows the differences in the box below.
- **Retake Restore point** replaces it with the current files. The old one is kept as a save point; you are told if that takes you over the limit.

11.3 Restore

1. Pick a point in **From**: the Restore point, a save point, an automatic copy, or a preset.
2. Tick the systems to restore. Only those systems' files are written.
3. Press **Restore selected**. The current files are saved first as an automatic copy, and the output lists each file as written, unchanged or skipped.

Delete removes the selected save point or automatic copy.

After any change, two cards appear:

- **Restart services** ("Config files changed"): buttons to restart exactly the services those files belong to, such as Asterisk, Allmon3, the DVSwitch bridges and usrp2m17. Nothing restarts until you press one.
- **Finish: reinstall the tools and reboot**: suggested so everything starts clean. It reinstalls the Node Controller tools from instmon's library, instmon first, then reboots. Each tool is saved first and put back if its new copy

doesn't start. Your callsign, node number and settings are kept. The node reboots only once instmon, the Dashboard and Asterisk are healthy. **Not now** hides the card.

> **CAREFUL** While the reinstall runs, don't power the node off. The page says when it is reinstalling, rebooting and done, and reconnects by itself.

11.4 Create save point

Type a name (up to 40 characters: letters, numbers, spaces, `_ . -`) and press **Create save point**. It saves the current files of every installed system.

> **TIP** Make it a habit: take a save point before editing files in SysMon, before an update, and before trying a new reflector setup.

11.5 Travel Node preset

Captures only this node's **wiring**:

- the private nodes (1000 to 1999) in `rpt.conf` and their `[nodes]` lines;
- their extensions, and `chan_usrp` in `modules.conf`;
- the `USRP2M17.ini` ports and the Analog_Bridge `[USRP]` ports. The digital-mode ports change with the mode, so they aren't included.

Never callsigns, IDs or passwords. **Preview** shows what would be saved; **Save as Travel Node** stores it. To use it, pick it in Restore on any node. Applying it also keeps Allmon3 in step with the private nodes, and gives bridge nodes (USRP channels) the standard USRP values. If the preset would remove a private node that EchoLink is attached to, it warns you first.

11.6 Identity

A check that your callsigns, IDs and passwords agree across every file that holds a copy. Values are masked (`••••••`). **Reveal** asks for the root password, then shows everything, passwords included, until you press **Hide** or reload.

BAND	GROUPS	RULE
Same everywhere	Callsign; ASL node number; DMR ID (7 digits)	Every copy should hold the same value.
Must match	DMR ID + suffix; AMI secret; ASL node password; BrandMeister password	Copies that are paired must agree, or the link fails.
Stands alone	values with no partner	Shown for reference only.
Phone accounts	the Dashboard's phone networks	Change these on the Dashboard's Phone tab.

Each group shows a chip: **matches**, **differs** (red), **some copies empty**, **not set**, **no public node**, or **one copy**. 44helper's own callsign and node (`44helper.conf`) are no longer listed.

Set the callsign everywhere TESTING

The Callsign line has a text box and **Set everywhere**. Type the new callsign: the box shows capitals and accepts 3 to 7 letters and digits, with at least one digit. Set everywhere first shows a preview of every line it will change, old and new; press OK to write.

- It writes every callsign field the card lists: the `rpt.conf` ID recording and ID talkover (`|i` text only; a sound file is left alone), EchoLink (its `-L` / `-R` stays), MMDVM_Bridge, `var.txt`, ircDDBGateway, the YSF, P25 and NXDN gateways and USRP2M17 (the part after the callsign stays). It also writes the Dashboard's `/etc/asl_dvs/asl_dvs.conf` and 44helper's own config. Each line keeps its format; files that are not on the node are listed as skipped.
- The ircDDB and DPlus logins change too. Those accounts must be registered under the new callsign, or the D-Star gateway will not log in.
- An automatic save point is taken first, so Restore can put the files back.
- Nothing is restarted. The report lists the services to restart. Restart the Dashboard (`nodecontrol`) first: until then it holds the old callsign, and its next Save or M17 connect puts the old callsign back.
- DMR IDs, node numbers and passwords are still changed in their files, or for phone accounts on the Dashboard's Phone tab.

11.7 Digital-mode gateways

Shows the gateway services behind D-STAR and XLX (ircDDBGateway), YSF and FCS, P25, NXDN and STFU. P25 and NXDN are installed but disabled, so out of the box they only start once you open their Dashboard page. **Enable and start at boot** makes every mode ready right after boot. **Undo** puts them back as they were.

11.8 Port check

Reads `rpt.conf`, Analog_Bridge, DVSwitch.ini, MMDVM_Bridge, USRP2M17, ircDDBGateway and the YSF, P25 and NXDN gateways, and reports:

- any UDP port two programs both **listen** on. A sender pointed at a listener is a pair, not a clash;
- whether each gateway's ports mirror MMDVM_Bridge's, and DVSwitch.ini's RemotePort matches the gateway's command port.

Check what is listening now adds the live picture: which ports are reachable from the network, the gateway command ports (which should never be reachable), ports only open through a broad firewall range, and configured ports that nothing is listening on.

11.9 Common tasks

Undo a change that broke the node

1. Config tab, Restore: pick the save point or automatic copy from before the change.
2. Tick the system involved (usually ASL3 or DVSwitch) and press Restore selected.
3. In Restart services, restart what it lists, or run Finish: reinstall and reboot.

Copy a working travel setup to a new Pi

1. On the working node: Travel Node preset, Preview, then Save as Travel Node.

2. On the new node, which uses the same instmon library: install the software (Chapter 9), then Restore and pick Travel Node.
3. Identity: type this node's callsign on the Callsign line and press Set everywhere; fill in its IDs and passwords in their files until every group shows "matches".

Before asking for help

1. Identity: no red **differs** chips.
2. Port check: no clashes, and Check what is listening now is clean.

UNDER THE HOOD

Where the copies live and how restores are written

Library

- Everything lives under `/etc/asl_dvs/instmon_library/config_restore/` (or `$INSTMON_LIBRARY_DIR`), so instmon's Backup & Export ZIP includes it. Automatic copies go under `auto/`; each point's ID is `YYYY-MM-DD_HHMMSS_name`.
- Files per system: **ASL3**: `rpt.conf`, `iax.conf`, `extensions.conf`, `modules.conf`, `manager.conf`, `allmon3.ini`; optional `echolink.conf`, `rpt_http_registrations.conf`, `savenode.conf`, `simpleusb.conf` and the `/etc/asterisk/custom/` copies. **DVSwitch**: `MMDVM_Bridge.ini`, `DVSwitch.ini`, `Analog_Bridge.ini`; optional `dvsm.macro`, `var.txt`, `/etc/ircddbgateway`, the YSF, P25 and NXDN gateway ini files. **USRP2M17**: `USRP2M17.ini`, `/etc/sudoers.d/usrp2m17`, plus `rpt.conf`, `modules.conf` and `extensions.conf`. **Phone**: `/etc/asl_dvs/phone.json`; optional `dvs_phone_*.conf` and `pjsip.conf`.
- A system counts as installed when its marker exists: `rpt.conf`, the MMDVM_Bridge or Analog_Bridge program, `/opt/USRP2M17`, or `phone.json`.

Writing a restore

- Each file is written through a `.44htmp` temporary file, created mode 600 and with `O_NOFOLLOW`, then given the saved owner and mode and swapped into place. Identical files are left alone ("unchanged"). Files whose folder no longer exists are "skipped".
- A sudoers file must pass `visudo -c` before it is put in place.
- Blocks managed by another tool, between `; >>> … managed by … >>>` and `; <<< … <<<` (such as the Dashboard's Phone block), are recognised when a preset is applied.

Identity scan

- Fields come from `rpt.conf`, `iax.conf`, `rpt_http_registrations.conf`, `manager.conf`, `echolink.conf`, the DVSwitch ini files, `var.txt`, `/etc/ircddbgateway` and `phone.json`. In ini files a `;` is part of the value; in Asterisk files it starts a comment.
- Callsigns are compared without a `-L`/`-R` suffix. DMR IDs are compared on their first 7 digits for the "same everywhere" band. Only node numbers of 4 digits or more are masked.
- Reveal uses the same root-password check and lockout as the login. Values are never written to the log.

Reinstall job

Runs as its own systemd unit, `asl-dvs-reinstall`, so it survives 44helper restarting itself. Its state and log are in `config_restore/reinstall_job.json` and `reinstall.log`. It uses a 90-second health guard while the tools come back up.

PART IV

The other tools

The Dashboard is where you operate the node day to day; instmon installs and updates the suite; wifimon keeps the Pi on the network and shuts it down safely when the power or WiFi fails.

12. The Dashboard
13. instmon
14. wifimon and wifi_menu.sh

The Dashboard

The Dashboard (Node Control) is the page you use on the air: link AllStar and EchoLink nodes, pick DMR talkgroups and reflectors on every digital mode, make phone calls, and keep favorites. Open it at `http://<pi-address>:8989`. The Pi Zero 2 W build has every tab of the full build.

12.1 The page



Figure 12.1 · Illustration. The DMR tab. Every mode tab has the same shape: a quick bar, an optional search card, a grid of favorites, and Disconnect buttons.

PART	WHAT IT DOES
Header	Callsign, node number, uptime, Log Out, version.
Quick links	Allmon3, the DVSwitch page, Cockpit, instmon, wifimon and SysMon. SysMon is greyed out while it isn't running.
Indicator	What you are connected to now (talkgroup, node or reflector). The dots light while traffic passes.
Quick bar	Type a number or reflector and press **Tune** (or Enter). **LH** opens that network's Last Heard page; **Save** adds it to your favorites.
Search card	On most tabs: search the full list (AllStar nodes, EchoLink stations, reflectors, talkgroups), pick one, then **Connect**, **LH** or **Save**.
Favorites grid	One tap to connect. The one in use is highlighted in green, and its name turns gold while the radio is keyed.
Disconnect / Disconnect All	Drop this tab's connection, or every connection on the node.
System	**Reboot** and **Shutdown** for the Pi, at the bottom of the page (both ask first).

> **NOTE** **Changing tab changes mode.** When you open a mode tab, the Dashboard drops the current links and talkgroup, switches DVSwitch to that mode, starts its gateway if needed, and brings up the bridge node it uses. Leaving the Phone tab hangs up any call; leaving M17 disconnects the reflector. So the tab you are on is the mode you are on.

12.2 The tabs

TAB	CONNECT TO	NOTES
ASL	AllStarLink nodes	Type a node number, or search by node, callsign or place. Disconnect drops the current link.
Echo	EchoLink stations	Node number or callsign search.
DMR	Talkgroups on a DMR network	Pick the network (BrandMeister, TGIF, DMR+...) in the drop-down first; each network has its own favorites.
STFU	BrandMeister talkgroups or private IDs, directly	Uses the STFU terminal (Chapter 4), not MMDVM_Bridge's DMR.
YSF	System Fusion reflectors	Filter by country and name, or type `IP:port`.
FCS	FCS rooms	Pick a server, then a room, or type `FCS00335` or just `335`.
P-25 / NXDN	Talkgroups	Search by number or address.
D-STAR	REF, XRF and DCS reflectors, or a gateway callsign	Pick the module letter beside the reflector. Linking to a gateway callsign is **TESTING** (below).
XLX	XLX reflectors	Reflector plus module.
M17	M17 reflectors	Reflector plus module (or All Modules), or type the reflector IP.
Phone **WORKING**	Telephone calls	Below. Hams Over IP is **TESTING**.
EDIT	Settings and favorites	Below.

CAREFUL A red **PERMANENT LINK ACTIVE** bar means your node has a permanent link to another node. Anything you connect on a digital mode is then bridged onto that link too. Press **Disconnect** in the bar unless you mean to bridge.

TIP "⚠ Tune Failed" pop-ups say why: the reflector didn't answer, a gateway service didn't start, and so on. A gateway that won't start shows in SysMon's Services tab.

Linking to a D-STAR gateway TESTING

TESTING Linking the node to another D-STAR gateway (a repeater or a hotspot by its callsign) is still being tested. Reflectors (REF, XRF, DCS) are not affected.

Besides reflectors, the D-STAR tab can link your node straight to another D-STAR gateway, such as a repeater:

1. Type the gateway's callsign in the D-STAR box, for example `W1ABC`. It is 3 to 6 letters and digits with at least one digit. A name made of REF, XRF or DCS plus 3 digits is always taken as a reflector.
2. Pick the module letter of the far repeater you want, usually **A**, **B** or **C**. You can also type it after the callsign, as in `W1ABC B`.
3. Press Tune. The pop-up says *D-STAR link sent to gateway W1ABC B*. That means the link request went out, not that the far end accepted it, so listen or check the far gateway's dashboard.
4. **Disconnect** unlinks it.

What has to be in place, in `/etc/ircddbgateway`:

SETTING	WHY
`ircddbEnabled=1`	Required. The far gateway is found by its callsign on ircDDB, so it must be registered there too and have the module you picked.
`dextraEnabled=1`	Gateway links use DExtra.
`gatewayCallsign`, `repeaterBand1`	Your gateway's callsign and module letter. Together they name your node, such as `N0CALL B`.
`remoteEnabled=1`, `remotePassword`, `remotePort`	Optional, but recommended. With ircDDBGateway's remote control on, the Dashboard asks ircDDBGateway directly to link your node to the far gateway and module, and hears back whether the link was accepted, so a refused link shows as an error. It also works when ircDDBGateway's reconnect is set to a fixed link, which ignores ordinary link commands. Without it, or if remote control doesn't answer, the Dashboard falls back to tuning through `dvswitch.sh` as it does for a reflector.

SysMon's **D-Star — ircDDBGateway** card (section 4.2) checks every one of these and says what to fix.

> **CAREFUL** The remote-control port lets anyone who knows the password link and unlink your gateway. Keep it on the Pi: `gatewayAddress` blank or the Pi's own address, and never forward the port on your router.

12.3 Phone tab WORKING

Once the phone system is set up and activated on the EDIT page, the Phone tab places and answers calls through Asterisk.

> **WORKING** The phone system works and has been used on a live node.

> **TESTING** **Hams Over IP** as the phone network has been used to make calls, but it is still being tested. If something goes wrong, press **Report** on the Phone tab and include the text when you ask for help.

CONTROL	WHAT IT DOES
Network drop-down	Which phone network to use. Only the picked network signs in, so two accounts for the same number never compete. The dot shows sign-in state.
Number + Dial	Places the call. It joins your node when answered.
***99 / #**	Send the "talk" and "unkey" tones into the call.
Favorites	Numbers you saved, plus Test call and Voicemail rows.
Hang Up · Keypad · Patch · Report	End the call; a keypad for menus; phone patch on or off; a text report to copy when asking for help.

- The phone node links to your radio node **only while the Phone tab is open**. With it closed, callers get busy.
- From the radio: `*61` + number dials (the Phone tab must be open), `*62` or `*65` hangs up, and `*980`, `*981` and `*982` send `*`, `#` and `*99`.
- Calls join with voice keying, so a simplex radio only transmits while the far end talks. Long nonstop talk from the far end is broken for 2 seconds every 10.

Why your radio isn't keyed for the whole call

AllStar has a built-in phone patch, the **autopatch**. While an autopatch call is up, AllStar keys every node linked to it for the whole call, whether anyone is talking or not. The code that would key only on voice is switched off inside AllStar itself, because it cut off the first word. No setting in `rpt.conf` changes this.

On a node with a simplex hotspot that is a **dead key**: the phone node keys its link to your radio node, your hotspot transmits nonstop, and it never hears you. You hear the caller but can't talk back, and `*99` seems to do nothing.

The Dashboard (9.3.71.8-pi02w and 9.3.73 on) doesn't keep the call in autopatch:

- **Calls you dial** (Dial, favorites, Test call, Voicemail) are placed by the Dashboard, and join the phone node with `rpt(<node>,Pv)` once the far end answers.
- **Incoming calls** join the same way.
- ***61 from the radio** still uses autopatch, but only to collect the number. The autopatch ends at once and the Dashboard places the call the same way. Your hotspot is keyed only for those few seconds of dialing.

How `Pv` works

- **P** (phone mode): the caller joins your node like another linked station. They hear everything on the node, and could key the radio with `*99`.
- **v** (voice keying, VOX): no `*99` needed. The caller's voice keys the transmitter.

1. The far end talks: AllStar detects voice and keys your hotspot.
2. The far end stops: after about 0.4 s of quiet the hotspot unkeys and goes back to receive.
3. You key your radio and talk: your audio goes into the call.
4. When the far end starts talking, about 0.5 s of audio is held back, so the transmitter is up before the first word arrives.
5. If the far end talks for 10 s without a break, the hotspot unkeys for 2 s so you can break in.

> **NOTE** The trade-offs: you don't hear ringing over the air (you hear the call once it's answered); there is about half a second of delay; nonstop far-end talk gets a 2-second gap every 10 seconds; and a very quiet caller might not trigger the voice keying.

12.4 EDIT page

SECTION	HOLDS
Configuration	Callsign; main node number; Digital Voice Bridge node (1999); M17 Bridge node (1917); the phone nodes (information only); web port (8989); CPUWeight.
Visible Tabs	Tick the mode tabs you use.
Phone	**Activate** (sets the phone system up in Asterisk, asks first); phone patch; `*61` digit timeout (how long `*61` from the radio waits for the next digit); restart Asterisk if needed; transmitter time-out off while the tab is open, with a safety cap in minutes; **Listen only** (you hear calls, but nothing from the radio goes into them); how keypad tones are sent; and the **Networks** list (SIP or IAX2 accounts, with Add network). **Save Phone**, **Reload**, and **Revert (turn phone off)**, which takes it all back out of Asterisk.
Favorites lists	Every mode's favorites, to rename, reorder or delete.

Save writes the settings and favorites; **Reload** throws away unsaved edits; **Back** returns to the tab you came from.

> **NOTE** Activate and Save Phone rewrite the Dashboard's own blocks in `rpt.conf`, `extensions.conf`, `pjsip.conf` / `iax.conf` and `modules.conf`, then reload Asterisk. Take a 44helper save point first (Chapter 11). Every detail is in `docs/ASL-DVS_Pi02w_Phone_Dialplan_Guide.pdf`.

12.5 Common tasks

Join a DMR talkgroup and save it

1. DMR tab: pick the network, type the talkgroup number and press Tune.
2. Check the indicator shows it, then press Save.

Link to an AllStar hub by name

1. ASL tab: type part of the hub's callsign or town in the search box.
2. Pick it in the list and press Connect. Use Disconnect when you're done.

Make a test phone call

1. Phone tab: pick the network and wait for the sign-in dot.
2. Press the Test call row. SysMon's Phone tab Router test (Chapter 4) checks the audio while the call is up.

UNDER THE HOOD

How the Dashboard drives the node

- **Settings:** `/etc/asl_dvs/asl_dvs.conf` (identity, favorites, visible tabs, DMR networks). Phone: `/etc/asl_dvs/phone.json` and the `dvs_phone_*.conf` files it writes.
- **Polling:** `/api/status` every 5 s when idle and every 1 s while a tune is in progress, with an ETag, so unchanged state costs almost nothing. On the Pi Zero 2 W build the AllStar and EchoLink lists are searched on disk rather than held in memory.
- **Tuning:** through DVSwitch's `dvswitch.sh` (mode and tune commands) and Asterisk's AMI on 127.0.0.1:5038 (link and unlink). A tune counts as done when Analog_Bridge's `/tmp/ABInfo_*.json` confirms it.
- **Changing tab:** drops the links except the needed bridge node, sends a talkgroup disconnect, switches the DVSwitch mode, starts or stops the mode's gateway service (YSF, P25, NXDN, ircDDBGateway), and links the bridge node (1999, or 1917 for M17).
- **M17:** writes `/opt/USRP2M17/USRP2M17.ini` in full on every connect and disconnect, and restarts `usrp2m17`. From 9.3.71.11 (full build 9.3.74) it listens on UDP 17010 (it was 32010, which clashed with P25).
- **Phone calls** are placed with AMI `Originate` into `rpt(<node>,Pv)`. Radio `*61` requests arrive through `/run/asl_dvs_tones/req` and are dialed within about 2 seconds.
- **The dead key in AllStar's source** (`apps/app_rpt.c`, AllStarLink app_rpt, master `d392975`): in the per-link keying loop, `mycalltx = myrpt->callmode` keys every link while an autopatch is up; the voice-gated line is inside `#ifdef DONT_USE__CAUSES_CLIPPING_OF_FIRST_SYLLABLE_ON_LINK`, so it is never built (lines 4916–4921). With `farenddisconnect=1` or a half-duplex node the patch drops as soon as the dialplan ends (line 1294), which is how `*61` hands over.
- **How `Pv` keys**: the `v` option sets phone VOX on the link (app_rpt.c lines 7354–7367). `dovox()` (`rpt_vox.c`) turns on after 3 voiced 20 ms frames and off after 20 quiet ones. While the caller's link is "receiving", the node keys its other links (lines 4898–4906 and 3567–3581). `simplexphonedelay` (25 × 20 ms) delays the audio, and `voxtimeout` / `voxrecover` (10000 / 2000 ms) force the break. These are the values the Dashboard writes for each phone node.
- **Service:** `nodecontrol` (`asl_dvs_dashboard` before 9.3.76 / 9.3.71.15-pi02w), with `Restart=always` and `WatchdogSec=30`, so it is restarted if it stops answering. That is why the separate Watchdog script isn't needed.
- The login is shared with SysMon (Chapter 3).

instmon

instmon (Install Monitor) installs, starts, stops, updates and removes the other tools; keeps a library of every version you have had; and makes whole-card backups and clones. Open it at `http://<pi-address>:8990`. It is one long page of sections, from top to bottom.

13.1 Quick Links

Buttons for Allmon3, the DVSwitch page, Cockpit, the USRP2M17 page and **Router**. Router finds your network's gateway and opens its web page, if you are on the same network as the Pi.

13.2 GitHub Updates

Check GitHub, Update, Full Update, the Pi Zero 2 W build choice and Quiet System are covered in Chapter 2 (section 2.4). In short:

- **Check GitHub** compares every tool with this repository and marks the newer ones.
- **Full Update** updates them all, one at a time, putting the old copy back if a new one fails, with instmon itself last.
- **Quiet System** pauses SysMon, the Dashboard, 44helper and the Watchdog timer for up to 30 minutes, or until you press **Restore**.

Each tool's library group also has its own GitHub row, so you can fetch one new version without updating the rest.

13.3 Components

One card per tool: Dashboard, SysMon, wifimon, 44helper, Watchdog and instmon.

PART	MEANING
Version and badge	The installed version, and **RUNNING**, **STOPPED**, **FAILED** or **UNKNOWN**.
Port	Its web port, and "(not listening)" if the page isn't answering.
Start · Stop · Uninstall · Go	Control the service; **Go** opens its page. instmon can't stop or uninstall itself from here.
Library (nested)	Press **Show** for every stored version of that tool (below).

> **CAREFUL** As the red banner says: **NO CONFIRMATIONS**. Stop and Uninstall act the moment you press them. Uninstalling the Dashboard takes Node Control away until you install it again.

13.4 The library

instmon keeps every file it has downloaded or you have uploaded, grouped by tool, plus **Scripts (.sh)** and **Configuration**. Each row shows its version, name, size and date, and a badge: **INSTALLED** (the copy in use), **ALT** (another version), or **READY**.

BUTTON	WHAT IT DOES
Install	Installs that version: runs its `--install`, replacing the one in use. Use it to go back to an older version.

Install Config	Puts a stored config file back in place (Configuration group).
Run Script / Install / Run Uninstall	For `.sh` files: runs it, as an installer or uninstaller if the name says so. Menu-driven scripts such as `wifi_menu.sh` show "Interactive — run via SSH" instead.
Edit · Download · Delete	Open it in an editor, save a copy to your computer, or remove it from the library.
Upload & stage	Add a file from your computer to that group. Nothing is installed until you press Install.

> **TIP** A new version misbehaving? Open that tool's library and press Install on the version you had before, marked ALT. Full Update also leaves the previous copy here.

Import from USB Drive

The card above the library copies files from a USB stick into it, for a Pi with no internet or no computer handy. Pick the drive, pick the library group (Scripts, Config, or a tool), open folders, and press **Import** beside a file. Only `.py`, `.sh`, `.conf` and `.pdf` files are listed. A `.pdf` goes to Manuals whatever group is picked (from 2.42.0). The rules are those of Upload & stage: the right extension, a name that looks like that tool's file (or say yes to "Import anyway?"), and the same size limit. **Up** goes back a folder; **Refresh** re-reads the drives. The stick is mounted read-only and nothing on it is changed (from 2.39.1 that includes an ext3/ext4 journal left by unplugging without ejecting). The boot disk is never listed, and the card waits while a disk-image job runs. From instmon 2.39.0.

Manuals

The Manuals card, below the library, keeps this guide and the other manuals (PDF) on the node. **Get from GitHub** downloads the files in the repo's `docs` folder, only the ones that are new or changed. Each manual then has **Open**, which shows it in a new browser tab in the browser's own PDF reader, and **Download**, which saves it to your computer. Every download is checked against GitHub's file list and must be a PDF before it is kept. From instmon 2.41.0. On a Pi with no internet, import the PDFs from a USB stick with Import from USB Drive instead (2.42.0 and later): spaces and other odd characters in the name become `_`, and the file must be a real PDF.

From 2.44.0 any PDF can be added from your computer too: pick it with **Upload PDF** on the Manuals card (or with any library group's **Upload & stage**; a `.pdf` always goes to Manuals). The same name and "real PDF" rules apply, up to the upload size limit (50 MB unless `INSTMON_MAX_UPLOAD_MB` says otherwise). Each manual has **Delete** as well as Open and Download. From 2.46.0 a manual from GitHub that you delete stays deleted through Full Update, and comes back only when you press Get from GitHub (or upload or import one with that name).

From 2.45.0 you don't need to press Get from GitHub on a new node. `--install` downloads the manuals, and if the node was offline then, instmon tries again when it starts (after 5 seconds, then 1, 2 and 5 minutes later) until one download works. After that first download it only fetches manuals on Get from GitHub and Full Update, both of which take only the files that are new or changed on GitHub. There is no setting to turn this off.

From 2.46.0 **Check GitHub** also reads the `docs` folder. Manuals that are new or changed there show in the Full Update list as **Manual NEW** or **Manual CHANGED**, and turn the Full Update button on even when every tool is current. A manual you deleted is not listed.

13.5 Backup & Export

Download Full Backup (ZIP) saves the whole library to your computer: every tool version, script and config. That includes 44helper's restore points and the WireGuard config copy kept there (Chapters 8 and 11), so keep the ZIP safe. The manuals are left out; Get from GitHub fetches them again.

13.6 Disk Image & Clone

Whole-drive jobs for the Pi's SD card, or a drive in a USB adapter. The top card checks the tools these jobs need and lists the attached drives, refreshing every 5 seconds. Each job card (folded until you press Show) previews the exact command, has its own console, and has **Stop** and **Reset**.

CARD	WHAT IT DOES	MAIN OPTIONS
Full Format	Wipes a *spare* drive and makes one partition (ext4, FAT32 or exFAT). Never offered for the boot disk.	Quick format: clears only the start, so it is fast but not a secure erase.
Backup	Copies this Pi's whole disk to an image file on another drive.	Label; Shrink to fit; Verify after copy (sha256 or blake2b); Encrypt (AES-256, passphrase); Pause node services; Freeze filesystem; Continue past read errors; Standard (dd) or Resilient (ddrescue); throttle 5, 10 or 20 MB/s. The card also lists existing backups and has **Scheduled Backups**: every N hours to a drive, keeping the newest N.
Clone	Copies this Pi's disk straight onto a spare drive, with no file in between.	Verify; Pause node services; Freeze; Continue past read errors; Direct I/O; dd or ddrescue; throttle.
Restore	Writes a backup image onto a spare drive, never the running disk.	Passphrase for encrypted images; Verify; throttle (unencrypted only).

CAREFUL Format, Clone and Restore erase the destination drive completely. Double-check the drive you pick, and unmount it first. **Pause node services** takes the node **off the air** for the whole job (SysMon keeps running so you can watch); a scheduled backup with it ticked goes off the air on its own, unattended. An encrypted backup can't be recovered without its passphrase.

NOTE Without Freeze, an image of a running Pi is "crash-consistent", like pulling the power: ext4 tidies up on the first boot. That is normally fine. For the cleanest image, tick Pause node services and Freeze filesystem.

13.7 System & Comms, Execution Log

- **Comms SERV Restart**: stops Asterisk, Analog_Bridge, MMDVM_Bridge, STFU and Allmon3, waits 3 seconds, then starts them all. It asks first, and every link and call drops.
- **Reboot Pi** / **Shutdown Pi**: each asks twice. After Shutdown, someone has to power-cycle the Pi.
- **Execution Log**: everything instmon has done this session, with **Copy Log**.

13.8 Common tasks

Go back to the previous SysMon

1. Components, SysMon card: press Show on its library.
2. Find the version marked ALT and press Install.

Load new versions from a USB stick

1. Copy the new files onto a stick on your computer, and plug it into the Pi.

2. Import from USB Drive: pick the stick and the tool's group, open the folder, press Import on each file.
3. On the tool's card, open the library and press Install on the new version.

Keep a spare SD card ready

1. Put a spare card in a USB adapter and plug it into the Pi.
2. Clone: pick the card, tick Verify after copy and Pause node services, then press Clone Now.
3. Label the card with the date. If the main card fails, swap them.

UNDER THE HOOD

instmon's files and jobs

- **Program:** `/usr/local/bin/instmon.py`, service `instmon`, port 8990. Its own login: cookie `instmon_session`, 12 hours, kept in memory, so restarting instmon logs everyone out and the page reloads to the login screen.
- **Library:** `/etc/asl_dvs/instmon_library/` (`$INSTMON_LIBRARY_DIR`), one folder per category: `dashboard`, `sysmon`, `wifimon`, `44helper`, `watchdog`, `instmon`, `scripts`, `config`. 44helper adds `config/wg0.conf` and `config_restore/`.
- **Install** runs the file's own `--install`; **Uninstall** runs `--uninstall`. A tool is only offered for install if its file supports `--install` (and `--uninstall`) and passes the download checks in Chapter 2.
- **Version** comes from the file's `VERSION`, `APP_VERSION` or `SCRIPT_VERSION` line, or, for `.sh` scripts, the title line `# name.sh vX.Y (date)`.
- **Router link:** reads the default route from `/proc/net/route`, and probes the gateway on http:80, then https:443, with a 1.5-second limit, cached for 30 seconds. It is only offered when your browser is on that same network.
- **USB import:** a drive not already mounted is mounted `ro,nosuid,nodev,noexec` (plus `noload` on ext3/ext4, so a journal is never replayed) under `/mnt/instmon-usbimport/<drive>` for each request and unmounted after (FAT, exFAT, ext2/3/4, NTFS with `ntfs3` or `ntfs-3g`). Paths that leave the drive, symlinks and hidden entries are refused, at most 500 entries per folder. Endpoints `/api/usbimport/drives`, `/browse` and `/import`.
- **Manuals:** kept in `/etc/asl_dvs/instmon_library/docs/`. Get from GitHub reads the `docs` folder through the GitHub API, skips a file whose git blob id matches the copy on the node, and keeps a download only if its blob id matches the list and it starts with `%PDF-` (64 MB limit). Open and Download are `/api/docs/file?name=…` (add `&download=1` to save), served as `application/pdf` behind the login and without the page's frame and script rules, which the browser's PDF reader doesn't need and can refuse to run under.
- **Disk jobs:** `dd` or `ddrescue`, `openssl enc` with PBKDF2 for encryption, `resize2fs` and `parted` for Shrink to fit, and `fsfreeze` for Freeze. Throttled copies run as a shell pipeline that won't survive an instmon restart.

wifimon and wifi_menu.sh

wifimon keeps a WiFi node connected, and shuts it down cleanly before low voltage or a lost network can corrupt the SD card. Its page manages your WiFi networks, a travel hotspot and its own watchdogs. Open it at `http://<pi-address>:8991`. `wifi_menu.sh` is a text menu for the same WiFi jobs over SSH.

14.1 Status

- The network you are on, a signal meter from −90 to −30 dBm, the IP address, and whether the internet answers.
- Warnings with one-click fixes: **Power save is on** (it can cause dropouts; turn it off for this network or all of them) and **No WiFi country code** (it can keep a Pi's WiFi off or limited).
- **More details**, **Reconnect now**, the **WiFi country code**, and **WiFi chip roaming**. Stopping roaming helps with a single router, and takes effect after a reboot.

14.2 Saved networks

Every saved WiFi network, in the order the node tries them. **Add network** takes a name, security (WPA2, WPA3 or open), password, a hidden-network box, **Use this Pi's real MAC address** (for hotels and routers that register devices) and power save. Then **Save & connect** or **Save only**.

- Each network has **Options**: automatic or fixed IP address, your own DNS servers, real MAC, and band (any, 2.4 GHz only, 5 GHz only).
- The **reconnect list** is the set of networks the WiFi watchdog will go back to. A network saved some other way shows "not in the watchdog's reconnect list" with an **Add** button. **Add all to reconnect list** does them all at once.

> **NOTE** **Changes that could cut you off undo themselves.** When you connect to a different network or change options, a banner counts down. If your browser can't reach the page on the new connection within 60 seconds (120 for options), wifimon puts the old settings back. Press **Undo now** to go back straight away.

14.3 Nearby networks

What the Pi can hear, with signal strength. **Rescan** looks again. Use Add network to save one.

14.4 Radio & device

- **WiFi radio** on or off, and the list of WiFi devices, for example a USB WiFi adapter beside the built-in one.
- **Emergency hotspot**: the node makes its own WiFi, so a phone can reach this page and add a hotel network. Set a name and password (**New** makes one), and when it turns off by itself (15, 30, 60 or 120 minutes). While it is on, choose which network to join afterwards, or press **Stop hotspot**. The node leaves its current WiFi while the hotspot runs.

14.5 Watchdog

WATCHDOG	DOES	DEFAULT
WiFi watchdog	Reconnects to the reconnect list. If there is no connection for the set time, it shuts the node down cleanly. It waits, with no timer, while no networks are saved.	shut down after 180 s offline; retry every 15 s
Voltage watchdog	Shuts the node down cleanly after a sustained low-voltage spell.	below 0.90 V core for 30 s

Two bars count up while a problem lasts: **No connection** and **Low voltage**. After you change networks, a grace period keeps the WiFi watchdog from acting while the new connection settles. **Change settings** opens:

SETTING	RANGE
Shut down after no connection for	60 to 86,400 s
Shut down after low voltage for	10 to 3,600 s
Low voltage means core below	0.70 to 1.00 V
Retry WiFi every ... while offline	5 to 600 s
Check the connection every	2 to 60 s
Internet check pings	address to ping
Detect login pages	off by default; checks for hotel sign-in pages (fetches a test page about every 5 minutes)
Keep WiFi power save off	off by default; when on, keeps it off on every network

Reset to defaults puts every setting back.

> **CAREFUL** On a new install both watchdogs start **off**, and the page asks "Turn the watchdogs on?" at your first login (Chapter 2). Add your networks first. When a watchdog shuts the node down, SysMon and the Dashboard show a "wifimon — Node Shutting Down" notice. Afterwards the Pi needs power cycling, or its WiFi back.

14.6 Service and Activity log

- **Service**: wifimon's version and state, **Restart service**, **Log out**.
- **Activity log**: the last 100 events, newest first, with a "warnings and errors only" filter and Copy. The full history is in `journalctl -u wifimon`.
- **Shutdown reports**: one is saved every time wifimon shuts the node down, and after any stop that wasn't clean (power lost or a crash). View or Download them; **Save a test report** makes one now. They are the first place to look after an unexpected power-off.

14.7 wifi_menu.sh

A text menu for when the web page isn't reachable but SSH or a keyboard is. The file is `wifi_menu_v3_1_20261011.sh` (version 3.1); run it from the folder it is in:

```
sudo bash wifi_menu_v3_1_20261011.sh
```

KEY	DOES
1	Refresh status
2 / 3	Turn the WiFi radio on / off
4	Switch the active WiFi device
5	Disconnect the active WiFi
6	Connect to a network on the default device, from a scan list; `h` for a hidden network
7	List saved connections; `f <#>` forgets one (asks first)
q	Quit

CAREFUL Turning the radio off or disconnecting over a WiFi SSH session cuts that session. Use a keyboard and screen, or a wired link, for those.

14.8 Common tasks

Set up at a hotel

1. Reach wifimon once, for example through your phone's hotspot saved as a network at home, and press **Start hotspot** under Radio & device.
2. Join the hotspot from your phone and open `http://<node>:8991`.
3. Add the hotel network, with **Use this Pi's real MAC address** if the hotel registers devices, and turn on Detect login pages if it has a sign-in page.
4. Choose it under "Afterwards connect to", then Stop hotspot.

Find out why the node switched itself off

1. Activity log, Shutdown reports: open the newest one. It says which watchdog fired, with the voltage and connection history before it.
2. For low voltage: use a better power supply and a shorter, thicker cable. SysMon's Pwr field (Chapter 3) shows under-voltage since boot.

UNDER THE HOOD

How wifimon works

- **Files:** program `/usr/local/bin/wifimon.py`, service `wifimon`, port 8991. Settings and the reconnect list are in `/etc/wifimon/wifimon.conf`. NetworkManager's own profiles are in `/etc/NetworkManager/system-connections`. Shutdown state, which SysMon and the Dashboard read, is in `/run/wifimon/shutdown_state.json`. Reports are in `/var/log/wifimon`; the newest 10 are kept, each up to 256 KB.
- **WiFi:** everything goes through NetworkManager (`nmcli`). wifimon gives NetworkManager a 30-second head start to reconnect by itself, counts 3 failed checks as "lost", and treats a link with no traffic for 60 s as stuck. It reorders NetworkManager's join order to match the reconnect list.
- **Undo:** a risky change is recorded as pending, with a 60-second deadline (120 for options). The page checks in from the new connection to keep it; otherwise the change is undone. A 180-second grace period follows each change.
- **Voltage:** sampled through the Pi firmware. Snapshots are taken every 10 s, or every 5 s while voltage is low; the last 20 go into a shutdown report.
- **Login:** its own, with the root password, checked through PAM. Sessions last 12 hours. The page polls status every 5 seconds.
- **Setup from the shell:** `sudo python3 /usr/local/bin/wifimon.py --setup-wifi` adds or imports networks. `wifi_menu.sh` uses `nmcli` only and doesn't touch wifimon's reconnect list.

PART V

Reference

What to do when something goes wrong, and the facts you'll want to look up: ports, files, services and terms.

15. Troubleshooting and recovery

Appendices A–E: port map, files and paths, services, the launcher and memory savers, glossary.

Troubleshooting and recovery

Recipes for the problems people actually hit, each going from the quickest check to the bigger fix. Before any fix that changes files, take a save point in 44helper's Config tab (Chapter 11). It costs nothing and makes every step below undoable.

15.1 Where to look first

SYMPTOM	FIRST STOP
Anything at all	SysMon Overview: red dots, `NR:` counts, the Pwr field (Chapter 3)
A config looks wrong	SysMon ASL-DVS, DVSM, STFU and M17 tabs (Chapter 4); 44helper Config: Identity and Port check (Chapter 11)
Installs or updates fail	44helper Update tab, Package recovery (Chapter 10)
The Pi went off by itself	wifimon Shutdown reports (Chapter 14)
Remote access is broken	44helper Overview and Ports (Chapters 7 and 8)

15.2 A page doesn't load

1. Try another tool's page. If none load, the Pi or its network is down: check power and WiFi, and use `wifi_menu.sh` on a keyboard.
2. If only one is down, check it from SSH. Use the service and port from Appendix C:

```
systemctl status sysmon --no-pager
journalctl -u sysmon -n 30 --no-pager
ss -tlnp | grep 9999
```

3. Start or restart it from instmon's Components card, or with `sudo systemctl restart <service>`.
4. Still failing after an update? Install the previous version from instmon's library (ALT row, Chapter 13).

> **TIP** Login refused every time: set the root password again with `sudo passwd root`. After 5 wrong tries, the Dashboard and SysMon lock that address out for 15 minutes; wait it out.

15.3 No audio, or audio one way only, on digital modes

1. SysMon Overview: Asterisk, analog_bridge and mmdvm_bridge must all be green.
2. Dashboard: check the indicator shows the talkgroup or reflector you expect, and that you are on that mode's tab.
3. SysMon ASL-DVS tab: node 1999's RX and TX ports in `rpt.conf` must be Analog_Bridge's pair the other way round.
4. SysMon Ports, and 44helper Config, Port check: no `[DUP]` and no clash. A known one: M17's old LocalPort 32010 clashes with P25. The Dashboard uses 17010 from 9.3.71.11 (Pi02w) and 9.3.74 (full).

5. DMR only: SysMon ASL-DVS, check `MMDVM_Bridge.ini` has a real DMR ID and the network enabled, and STFU (if you use it) has its hotspot password.
6. On a Split Tunnel: AllStar working while DVSwitch and M17 fail usually means an unedited `AllowedIPs`. Use 44helper Split Tunnel, step 4b (Chapter 8).

15.4 Other nodes can't link to me

1. SysMon ASL-DVS: `rpt_http_registrations.conf` (or `iax.conf` register lines) has your node and a password. On the full build, the Reg tab shows the live registration.
2. 44helper Ports: IAX2 4569/udp must not say **needed-but-not-open**.
3. Behind a home router with no tunnel, the router must forward UDP 4569 to the Pi. With a 44Net tunnel, the 44NetConnect zone must allow `iax2` (Chapter 7, Firewall).
4. Two nodes behind one router need different IAX ports: 44helper Nodes tab (Chapter 8).

15.5 Something broke after a change or update

WHAT CHANGED	UNDO WITH
A config file (edited anywhere)	44helper Config, Restore, from the save point or automatic copy taken before it, then Restart services.
A Node Controller tool version	instmon, that tool's library, Install on the ALT (older) row.
Linux packages	44helper Update: steps 2 and 9 to 11, then re-check the tool tabs.
A SysMon Hardware option	Turn it back off on the same card, then reboot.
A 44helper System Optimization card	That card's Revert.
Memory savers suspected	Turn them off one at a time (Appendix D), starting with the pool cap, and restart the tools.

15.6 The Pi is slow or runs out of memory

1. `free -m`. A Pi Zero 2 W has about 415 MB usable, and zram swap is normal.
2. Check every tool runs through the launcher: `ps ax | grep asl_dvs_launch` should list each one. A tool started as plain `python3 tool.py` uses about twice the memory, so run its `--install` again.
3. Close pages you aren't using. Each open page polls its tool, and Cockpit's page costs about 27 MB while open.
4. SysMon Hardware: turn off the video driver and lower GPU memory (Chapter 5).
5. Turn off services you don't use, such as PackageKit, P25Gateway or Allmon3. The Pi Zero 2 W install manual lists them.

15.7 The node switched itself off

1. wifimon, Activity log, Shutdown reports: the newest report says which watchdog fired.
2. Low voltage: improve the supply and cable. SysMon's Pwr field shows `WARN` or `⚡ UV` when it happens.

3. No connection: check the reconnect list has every network the node might use, and lengthen "Shut down after no connection" if the WiFi is often slow to come back.
4. Report says "stop that wasn't clean": the power was cut or the Pi crashed. Check the SD card (Chapter 5).

15.8 An install or update keeps failing

1. Read the "NOT RUN" reason on the step. Masked services: use the card at the top of the tab. Broken packages: use the "What to do next" buttons.
2. 44helper Update: run "Is apt still running?", then Package recovery 1, 2 and 3, then 1 again. **Don't reboot** while recovery 1 lists problems.
3. Run the failed step again. If it still fails, Purge on that tab (with Repair, then purge if offered), read "Still left behind", and install again.

15.9 I lost remote access

- **After starting a tunnel:** on the home network, or with a keyboard and screen, use the tunnel tab's Purge, or `sudo systemctl disable --now wg-quick@wg0`.
- **After a firewall change:** a runtime-only test goes away with `sudo firewall-cmd --reload` or a reboot. On the full build, if ufw was turned on with no SSH rule: `sudo ufw allow 22/tcp` from a local console, or `sudo ufw disable`.
- **Banned by fail2ban:** the web pages still work. In 44helper, Update, Custom: `fail2ban-client set sshd unbanip <your-ip>`. Bans expire after 10 minutes anyway.
- **Cloudflare hostname not answering:** 44helper Cloudflare, Verify the connector, and check the Zero Trust dashboard shows it Healthy.

15.10 Phone calls fail

1. Dashboard Phone tab open? The phone node only links while it is.
2. SysMon Phone tab, Health checks: fix every FAIL. Autopatch modules: both features should be READY.
3. SysMon Phone, Router test: Going out, then Call audio during a Test call, then Coming in.
4. Dashboard Phone, Report: copy it for anyone helping you. The Phone dialplan guide in `docs/` has the full call flows.

15.11 The SD card

- SysMon SD Card: READ-ONLY root, kernel storage errors, or fsck errors mean the card is failing. Image or clone it now (instmon, Chapter 13) and replace it.
- To restore onto a new card: instmon Restore writes a backup image to a card in a USB adapter. Swap it in and boot.
- To reduce wear: 44helper System Optimization (Chapter 10).

15.12 Asking for help

Gather plain-text copies. Every tool has Copy buttons:

COPY	FROM
Overview / Services / Ports lists	SysMon (Chapter 3)
A service's journal	SysMon Journal pop-up, with a filter
Hardware report	SysMon Hardware
Security results	SysMon Security
All Asterisk or DVSwitch config files	SysMon Edit, ASL AllConf / DVS AllConf: **remove passwords first**
Phone report	Dashboard Phone tab
Execution log	instmon
Activity log, shutdown report	wifimon

CAREFUL Never post your `wg0.conf` private key, AMI secret, node password, BrandMeister hotspot password or phone account passwords. The SysMon and 44helper check cards only say whether these are set; AllConf output and raw files contain them in full.

UNDER THE HOOD

Recovery commands at a glance

```
# service state, log and port for any tool (Appendix C)
systemctl status <service> --no-pager
journalctl -u <service> -n 50 --no-pager
ss -tulpn | grep <port>

# reinstall a tool from its installed file
sudo python3 /usr/local/bin/sysmon.py --install
# SysMon's security checks without the web page
sudo python3 /usr/local/bin/sysmon.py --selftest

# package system health
pgrep -a 'apt|dpkg|unattended-upgr'
sudo dpkg --audit
sudo dpkg --configure -a

# Asterisk
sudo asterisk -rx "rpt show registrations"
sudo asterisk -rx "iax2 show peers"
sudo asterisk -rx "rpt xnode <node>"

# WiFi and power
nmcli device status
vcgencmd get_throttled
free -m; zramctl
```

Port map

The tools

PORT	TOOL	REACH
8989/tcp	Dashboard	LAN or VPN only
8990/tcp	instmon	LAN or VPN only
8991/tcp	wifimon	LAN or VPN only
9997/tcp	44helper	LAN only, never on 44Net
9999/tcp	SysMon	LAN or VPN only

The node

PORT	USED BY	REACH
4569/udp	IAX2, AllStar links	open to peers
5198–5199/udp	EchoLink / SvxLink	open to peers if used
1667/udp	VOTER / RTCM	open to peers if used
4568/udp	Ampersand hub IAX	open to peers if used
5038/tcp	Asterisk AMI	127.0.0.1 only
34001 / 32001	USRP: node 1999 ↔ Analog_Bridge	127.0.0.1 only
34008 / 32008	USRP: node 1917 ↔ USRP2M17	127.0.0.1 only
34012 / 32012	USRP: node 1918 ↔ Zello bridge	127.0.0.1 only
31100 / 31103	Analog_Bridge ↔ MMDVM_Bridge (DMR; other modes use their own 3xx00 ranges)	127.0.0.1 only
17010 → 17000	USRP2M17 → M17 reflectors	outbound
62031	BrandMeister (STFU)	outbound
6073 / 6074 / 6075	YSF / P25 / NXDN gateway command ports	127.0.0.1 only
54321/udp	ircDDBGateway remote control	never public
80/tcp	apache2 (Allmon3) or lighttpd (DVSwitch page)	LAN
8080/tcp	Ampersand web UI	LAN
9090/tcp	Cockpit	LAN

10000/tcp	Webmin (optional)	LAN
7844/udp	cloudflared	outbound only

The full pairing rules are in `docs/ASL-DVS_Idealized_Node_Configuration.pdf`, section 4.

Files and paths

PATH	WHAT	WRITTEN BY
`/usr/local/bin/asl_dvs_launch.py`	shared launcher	every tool's --install
`/usr/local/lib/asl_dvs/`	one folder per tool, each holding that tool's program (linked from /usr/local/bin)	every tool's --install
`/usr/local/lib/asl_dvs/nodecontrol/`	Dashboard program (9.3.76 and 9.3.71.15-pi02w and later; `dashboard/` in the versions before, and directly in `/usr/local/lib/asl_dvs/` before that)	Dashboard
`/usr/local/lib/asl_dvs/sysmon/`	SysMon program (6.13.78 and 6.13.67.12-pi02w and later; `/usr/local/lib/asl_dvs_sysmon/` before)	SysMon
`/usr/local/lib/asl_dvs/instmon/`	instmon program (2.40.0 and later)	instmon
`/etc/asl_dvs/instmon_library/docs/`	manuals (PDF) for the Manuals card	instmon
`/usr/local/lib/asl_dvs/wifimon/`	wifimon program (5.29 and later)	wifimon
`/usr/local/lib/asl_dvs/44helper/`	44helper program (0.0.180 and later; `/opt/44helper/` before, now a compatibility link)	44helper
`/etc/asl_dvs/asl_dvs.conf`	Dashboard settings and favorites	Dashboard, installer
`/etc/asl_dvs/phone.json`	phone networks	Dashboard
`/etc/asl_dvs/launch_no_*`	memory-saver off switches	you
`/etc/asl_dvs/instmon_library/`	tool versions, scripts, configs; `config/wg0.conf`; `config_restore/`	instmon, 44helper
`/etc/sysmon/sysmon.conf`	SysMon settings, pinned list, radio presets	SysMon
`/etc/sysmon/console.enable`	console switch (full build)	SysMon / you
`/etc/44helper/44helper.conf`	44helper settings, tunnels, router, peers	44helper
`/etc/44helper/terminal.enable`	terminal switch	44helper / you
`/etc/wifimon/wifimon.conf`	wifimon settings and reconnect list	wifimon
`/var/log/wifimon/`	shutdown reports	wifimon
`/run/asl_dvs/auth_session.json`	shared Dashboard and SysMon logins	Dashboard, SysMon
`/run/44helper/sessions.json`	44helper logins	44helper
`/etc/wireguard/wg0.conf`	44Net tunnel (mode 600)	44helper
`/etc/firewalld/zones/44NetConnect.xml`	44Net firewall zone	44helper
`/boot/firmware/config.txt`	video and GPU memory options (backup `config.txt.asl_dvs.bak`)	SysMon Hardware

`/etc/asterisk/`	ASL3 config	asl-menu, you, Dashboard (Phone), SysMon, 44helper
`/opt/Analog_Bridge, /opt/MMDVM_Bridge`	DVSwitch config	dvs, you, 44helper
`/opt/USRP2M17/USRP2M17.ini`	M17 bridge	Dashboard (each connect)
`/etc/allmon3/`	Allmon3	allmon3-passwd, you

Services

SERVICE	WHAT	PORT
nodecontrol	Dashboard (Restart=always, WatchdogSec=30; `asl_dvs_dashboard` before 9.3.76 / 9.3.71.15-pi02w)	8989
sysmon	SysMon	9999
instmon	instmon	8990
wifimon	wifimon	8991
44helper	44helper	9997
asterisk	ASL3 / app_rpt	4569/udp
allmon3	Allmon3	
analog_bridge · mmdvm_bridge · md380-emu	DVSwitch	
ysfgateway · p25gateway · nxdngateway · ircddbgatewayd	digital-mode gateways	
stfu	BrandMeister terminal	
usrp2m17	M17 bridge	
asl-zello-bridge	Zello bridge (full-build SysMon tab)	
wg-quick@wg0 (or 44net-tunnel)	44Net tunnel	
firewalld	firewall	
cloudflared	Cloudflare Tunnel	
cockpit.socket	Cockpit	9090
svxlink / svxlink-usrp	SvxLink	
amp-server / amp-hub	Ampersand	
asl-dvs-reinstall	44helper Config reinstall job (temporary)	

```
# start · stop · restart · at boot
sudo systemctl start|stop|restart <service>
sudo systemctl enable|disable --now <service>
```

The launcher and memory savers

Every tool's service starts it through `/usr/local/bin/asl_dvs_launch.py`, which loads the tool as a module so Python keeps the compiled copy in `__pycache__`:

TOOL	STARTED DIRECTLY	THROUGH THE LAUNCHER
SysMon (Pi02w)	54 MB	26 MB
Dashboard (Pi02w)	45 MB	25 MB
instmon	34 MB	21 MB
wifimon	36 MB	22 MB
44helper	48 MB	26 MB

Launcher version 2 adds three more savers, each with an off switch:

SAVER	WHAT IT DOES	OFF SWITCH
`python3 -OO`	Drops built-in help text from the loaded code.	`/etc/asl_dvs/launch_no_optimize`
`MALLOC_ARENA_MAX=2`	At most 2 memory pools instead of up to 8 per CPU core.	`/etc/asl_dvs/launch_no_arena_cap`
`malloc_trim`	1 minute after start, then every 5 minutes, hands freed memory back to Linux.	`/etc/asl_dvs/launch_no_trim`

```
# turn one off for every tool, then restart them
sudo touch /etc/asl_dvs/launch_no_arena_cap
sudo systemctl restart sysmon nodecontrol instmon wifimon 44helper
# delete the file and restart to turn it back on
```

With `-OO`, the first start after an update compiles everything once more, so measure memory after the next restart. Each tool writes the launcher when it installs, and removes it on uninstall once no service uses it. The launcher's text is identical in every tool.

Glossary

TERM	MEANING
44Net / AMPRNet	The 44.0.0.0/9 and 44.128.0.0/10 address space for amateur radio. **44Net Connect** is ARDC's free WireGuard tunnel service that gives you one of these addresses.
Allmon3	ASL3's web page for watching node links.
AMBE	The voice codec used by DMR, D-STAR and Fusion. Done in software, or by a USB dongle such as a ThumbDV.
AMI	Asterisk Manager Interface (port 5038): how the Dashboard drives Asterisk. Keep it local.
Analog_Bridge / MMDVM_Bridge	The two DVSwitch programs that turn node audio into digital voice and connect it to the networks.
app_rpt	The Asterisk module that makes a repeater or AllStar node.
ASL3	AllStarLink version 3.
Autopatch / reverse autopatch	The radio dialing out to a phone / a phone calling in to the radio.
Bridge node	A private node (1000–1999) that links Asterisk to a bridge program, such as 1999 for DVSwitch or 1917 for M17.
Build: Pi02w / full	The two builds of the Dashboard and SysMon (Chapter 1).
Cockpit	A general web admin console for the Pi (port 9090).
DVSM	DVSwitch Mobile, the phone app.
firewalld / ufw	The two Linux firewall front ends the suite works with.
IAX2	Asterisk's protocol for node-to-node links (UDP 4569).
Launcher	`asl_dvs_launch.py`, which starts every tool with less memory (Appendix D).
Masked	A systemd service blocked from starting at all.
NR	A service's restart count since it was started.
Quiet mode / Quiet System	Pausing services before big jobs: 44helper's (until turned off) and instmon's (up to 30 minutes).
Restore point / save point	44helper Config copies of your config files (Chapter 11).
STFU	DVSwitch's direct BrandMeister terminal.
Split / Full tunnel	A 44Net tunnel carrying only 44Net traffic, or all traffic.
USRP	A simple UDP audio protocol between Asterisk and bridge programs. It has no password, so keep it on 127.0.0.1.
USRP2M17	The bridge between a node and M17 reflectors.
wg0	The WireGuard tunnel interface.

zram	Compressed swap kept in memory, used on the Pi Zero 2 W.

Node Controller · KD8PGK · written with Claude (Anthropic) · CC BY-NC 4.0 · github.com/kd8pgk/ASL-DVS. Change history for every tool is in the repository's `changelogs/` folder.